FoodAccessPREP

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2026-08-19

Background and Data

Our group used the 2015 and 2019 Food Access Research Atlas datasets, available through the U.S. Department of Agriculture Economic Research Service.

The datasets are of interest because they provide insight into access to supermarkets and essential food resources across different communities. They can be used to examine food access among people receiving benefits, those living in areas of high deprivation or low socioeconomic status, those living further away in urban vs rural areas, and different demographic groups. Using data from multiple years also allows us to identify changes over time and determine whether there have been noticeable improvements in food access.

The datasets contain information including census tract, poverty rate, food access at specified distances, and access measures across different racial and age groups, as well as SNAP beneficiaries. The majority of the variables are numerical, with some categorical variables such as census tract, state, and county. The data is structured so that individual variables represent specific measures of food access for particular demographic groups and distance thresholds.

One point of interest was that the 2019 dataset contained a large number of NULL values. These were assumed to represent zero values, as many of the affected variables represented whether an observation belonged to a particular demographic or access category. For example, a variable could represent the number of Native American individuals experiencing low food access at 20 miles, where a NULL indicated that there were no individuals meeting that condition. In contrast, both datasets also contained a considerable number of genuine NA values, particularly for SNAP_la_rate (2,405) and senior_la_rate (520).

To enable comparisons between years, our group used both the 2015 and 2019 Food Access Research Atlas datasets. The datasets were straightforward to integrate because they contained consistent and largely homogeneous variable names. We therefore retained the same variables from each dataset, allowing us to make meaningful comparisons in food access between 2015 and 2019.

DATA 2015

```
library(readxl)
DATA2015 <- read_excel("FoodAccessResearchAtlasData2015.xlsx", sheet = 3, na = c("", "NULL"))
head(DATA2015)
```

```
## # A tibble: 6 x 147
##   CensusTract State   County  Urban POP2010 OHU2010 GroupQuartersFlag NUMGQTRS
##   <chr>       <chr>   <chr>   <dbl>   <dbl>   <dbl>             <dbl>    <dbl>
## 1 01001020100 Alabama Autauga     1    1912     693                 0        0
## 2 01001020200 Alabama Autauga     1    2170     743                 0      181
## 3 01001020300 Alabama Autauga     1    3373    1256                 0        0
## 4 01001020400 Alabama Autauga     1    4386    1722                 0        0
## 5 01001020500 Alabama Autauga     1   10766    4082                 0      181
## 6 01001020600 Alabama Autauga     1    3668    1311                 0        0
```

```
## # i 139 more variables: PCTGQTRS <dbl>, LILATracts_1And10 <dbl>,
## #   LILATracts_halfAnd10 <dbl>, LILATracts_1And20 <dbl>,
## #   LILATracts_Vehicle <dbl>, HUNVFlag <dbl>, LowIncomeTracts <dbl>,
## #   PovertyRate <dbl>, MedianFamilyIncome <dbl>, LA1and10 <dbl>,
## #   LAhalfand10 <dbl>, LA1and20 <dbl>, LATracts_half <dbl>, LATracts1 <dbl>,
## #   LATracts10 <dbl>, LATracts20 <dbl>, LATractsVehicle_20 <dbl>,
## #   LAPOP1_10 <dbl>, LAPOP05_10 <dbl>, LAPOP1_20 <dbl>, LALOWI1_10 <dbl>, ...
```

DATA 2019

```r
DATA2019 <- read_excel("FoodAccessResearchAtlasData2019.xlsx", sheet = 3, na = c("", "NULL"))
head(DATA2019)
```

```
## # A tibble: 6 x 147
##   CensusTract State   County    Urban Pop2010 OHU2010 GroupQuartersFlag NUMGQTRS
##   <chr>       <chr>   <chr>     <dbl>   <dbl>   <dbl>             <dbl>    <dbl>
## 1 01001020100 Alabama Autauga ~     1    1912     693                 0        0
## 2 01001020200 Alabama Autauga ~     1    2170     743                 0      181
## 3 01001020300 Alabama Autauga ~     1    3373    1256                 0        0
## 4 01001020400 Alabama Autauga ~     1    4386    1722                 0        0
## 5 01001020500 Alabama Autauga ~     1   10766    4082                 0      181
## 6 01001020600 Alabama Autauga ~     1    3668    1311                 0        0
## # i 139 more variables: PCTGQTRS <dbl>, LILATracts_1And10 <dbl>,
## #   LILATracts_halfAnd10 <dbl>, LILATracts_1And20 <dbl>,
## #   LILATracts_Vehicle <dbl>, HUNVFlag <dbl>, LowIncomeTracts <dbl>,
## #   PovertyRate <dbl>, MedianFamilyIncome <dbl>, LA1and10 <dbl>,
## #   LAhalfand10 <dbl>, LA1and20 <dbl>, LATracts_half <dbl>, LATracts1 <dbl>,
## #   LATracts10 <dbl>, LATracts20 <dbl>, LATractsVehicle_20 <dbl>,
## #   LAPOP1_10 <dbl>, LAPOP05_10 <dbl>, LAPOP1_20 <dbl>, LALOWI1_10 <dbl>, ...
```

```r
dim(DATA2015)
```

```
## [1] 72864   147
```

```r
dim(DATA2019)
```

```
## [1] 72531   147
```

```r
names(DATA2015)
```

```
##   [1] "CensusTract"          "State"                "County"
##   [4] "Urban"                "POP2010"              "OHU2010"
##   [7] "GroupQuartersFlag"    "NUMGQTRS"             "PCTGQTRS"
##  [10] "LILATracts_1And10"    "LILATracts_halfAnd10" "LILATracts_1And20"
##  [13] "LILATracts_Vehicle"   "HUNVFlag"             "LowIncomeTracts"
##  [16] "PovertyRate"          "MedianFamilyIncome"   "LA1and10"
##  [19] "LAhalfand10"          "LA1and20"             "LATracts_half"
##  [22] "LATracts1"            "LATracts10"           "LATracts20"
##  [25] "LATractsVehicle_20"   "LAPOP1_10"            "LAPOP05_10"
##  [28] "LAPOP1_20"            "LALOWI1_10"           "LALOWI05_10"
##  [31] "LALOWI1_20"           "lapophalf"            "lapophalfshare"
##  [34] "lalowihalf"           "lalowihalfshare"      "lakidshalf"
##  [37] "lakidshalfshare"      "laseniorshalf"        "laseniorshalfshare"
##  [40] "lawhitehalf"          "lawhitehalfshare"     "lablackhalf"
##  [43] "lablackhalfshare"     "laasianhalf"          "laasianhalfshare"
##  [46] "lanhopihalf"          "lanhopihalfshare"     "laaianhalf"
```

```
## [49] "laaianhalfshare"     "laomultirhalf"        "laomultirhalfshare"
## [52] "lahisphalf"          "lahisphalfshare"      "lahunvhalf"
## [55] "lahunvhalfshare"     "lasnaphalf"           "lasnaphalfshare"
## [58] "lapop1"              "lapop1share"          "lalowi1"
## [61] "lalowi1share"        "lakids1"              "lakids1share"
## [64] "laseniors1"          "laseniors1share"      "lawhite1"
## [67] "lawhite1share"       "lablack1"             "lablack1share"
## [70] "laasian1"            "laasian1share"        "lanhopi1"
## [73] "lanhopi1share"       "laaian1"              "laaian1share"
## [76] "laomultir1"          "laomultir1share"      "lahisp1"
## [79] "lahisp1share"        "lahunv1"              "lahunv1share"
## [82] "lasnap1"             "lasnap1share"         "lapop10"
## [85] "lapop10share"        "lalowi10"             "lalowi10share"
## [88] "lakids10"            "lakids10share"        "laseniors10"
## [91] "laseniors10share"    "lawhite10"            "lawhite10share"
## [94] "lablack10"           "lablack10share"       "laasian10"
## [97] "laasian10share"      "lanhopi10"            "lanhopi10share"
## [100] "laaian10"           "laaian10share"        "laomultir10"
## [103] "laomultir10share"   "lahisp10"             "lahisp10share"
## [106] "lahunv10"           "lahunv10share"        "lasnap10"
## [109] "lasnap10share"      "lapop20"              "lapop20share"
## [112] "lalowi20"           "lalowi20share"        "lakids20"
## [115] "lakids20share"      "laseniors20"          "laseniors20share"
## [118] "lawhite20"          "lawhite20share"       "lablack20"
## [121] "lablack20share"     "laasian20"            "laasian20share"
## [124] "lanhopi20"          "lanhopi20share"       "laaian20"
## [127] "laaian20share"      "laomultir20"          "laomultir20share"
## [130] "lahisp20"           "lahisp20share"        "lahunv20"
## [133] "lahunv20share"      "lasnap20"             "lasnap20share"
## [136] "TractLOWI"          "TractKids"            "TractSeniors"
## [139] "TractWhite"         "TractBlack"           "TractAsian"
## [142] "TractNHOPI"         "TractAIAN"            "TractOMultir"
## [145] "TractHispanic"      "TractHUNV"            "TractSNAP"
```

```r
names(DATA2019)
```

```
##   [1] "CensusTract"         "State"                "County"
##   [4] "Urban"               "Pop2010"              "OHU2010"
##   [7] "GroupQuartersFlag"   "NUMGQTRS"             "PCTGQTRS"
##  [10] "LILATracts_1And10"   "LILATracts_halfAnd10" "LILATracts_1And20"
##  [13] "LILATracts_Vehicle"  "HUNVFlag"             "LowIncomeTracts"
##  [16] "PovertyRate"         "MedianFamilyIncome"   "LA1and10"
##  [19] "LAhalfand10"         "LA1and20"             "LATracts_half"
##  [22] "LATracts1"           "LATracts10"           "LATracts20"
##  [25] "LATractsVehicle_20"  "LAPOP1_10"            "LAPOP05_10"
##  [28] "LAPOP1_20"           "LALOWI1_10"           "LALOWI05_10"
##  [31] "LALOWI1_20"          "lapophalf"            "lapophalfshare"
##  [34] "lalowihalf"          "lalowihalfshare"      "lakidshalf"
##  [37] "lakidshalfshare"     "laseniorshalf"        "laseniorshalfshare"
##  [40] "lawhitehalf"         "lawhitehalfshare"     "lablackhalf"
##  [43] "lablackhalfshare"    "laasianhalf"          "laasianhalfshare"
##  [46] "lanhopihalf"         "lanhopihalfshare"     "laaianhalf"
##  [49] "laaianhalfshare"     "laomultirhalf"        "laomultirhalfshare"
##  [52] "lahisphalf"          "lahisphalfshare"      "lahunvhalf"
##  [55] "lahunvhalfshare"     "lasnaphalf"           "lasnaphalfshare"
```

```
##  [58] "lapop1"              "lapop1share"         "lalowi1"
##  [61] "lalowi1share"        "lakids1"             "lakids1share"
##  [64] "laseniors1"          "laseniors1share"     "lawhite1"
##  [67] "lawhite1share"       "lablack1"            "lablack1share"
##  [70] "laasian1"            "laasian1share"       "lanhopi1"
##  [73] "lanhopi1share"       "laaian1"             "laaian1share"
##  [76] "laomultir1"          "laomultir1share"     "lahisp1"
##  [79] "lahisp1share"        "lahunv1"             "lahunv1share"
##  [82] "lasnap1"             "lasnap1share"        "lapop10"
##  [85] "lapop10share"        "lalowi10"            "lalowi10share"
##  [88] "lakids10"            "lakids10share"       "laseniors10"
##  [91] "laseniors10share"    "lawhite10"           "lawhite10share"
##  [94] "lablack10"           "lablack10share"      "laasian10"
##  [97] "laasian10share"      "lanhopi10"           "lanhopi10share"
## [100] "laaian10"            "laaian10share"       "laomultir10"
## [103] "laomultir10share"    "lahisp10"            "lahisp10share"
## [106] "lahunv10"            "lahunv10share"       "lasnap10"
## [109] "lasnap10share"       "lapop20"             "lapop20share"
## [112] "lalowi20"            "lalowi20share"       "lakids20"
## [115] "lakids20share"       "laseniors20"         "laseniors20share"
## [118] "lawhite20"           "lawhite20share"      "lablack20"
## [121] "lablack20share"      "laasian20"           "laasian20share"
## [124] "lanhopi20"           "lanhopi20share"      "laaian20"
## [127] "laaian20share"       "laomultir20"         "laomultir20share"
## [130] "lahisp20"            "lahisp20share"       "lahunv20"
## [133] "lahunv20share"       "lasnap20"            "lasnap20share"
## [136] "TractLOWI"           "TractKids"           "TractSeniors"
## [139] "TractWhite"          "TractBlack"          "TractAsian"
## [142] "TractNHOPI"          "TractAIAN"           "TractOMultir"
## [145] "TractHispanic"       "TractHUNV"           "TractSNAP"
```

```
str(DATA2015)
```

```
## tibble [72,864 x 147] (S3: tbl_df/tbl/data.frame)
##  $ CensusTract         : chr [1:72864] "01001020100" "01001020200" "01001020300" "01001020400" ...
##  $ State               : chr [1:72864] "Alabama" "Alabama" "Alabama" "Alabama" ...
##  $ County              : chr [1:72864] "Autauga" "Autauga" "Autauga" "Autauga" ...
##  $ Urban               : num [1:72864] 1 1 1 1 1 1 1 0 0 0 ...
##  $ POP2010             : num [1:72864] 1912 2170 3373 4386 10766 ...
##  $ OHU2010             : num [1:72864] 693 743 1256 1722 4082 ...
##  $ GroupQuartersFlag   : num [1:72864] 0 0 0 0 0 0 0 0 0 0 ...
##  $ NUMGQTRS            : num [1:72864] 0 181 0 0 181 0 36 0 0 14 ...
##  $ PCTGQTRS            : num [1:72864] 0 0.0834 0 0 0.0168 ...
##  $ LILATracts_1And10   : num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
##  $ LILATracts_halfAnd10: num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
##  $ LILATracts_1And20   : num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
##  $ LILATracts_Vehicle  : num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
##  $ HUNVFlag            : num [1:72864] 0 0 0 0 1 0 1 1 0 0 ...
##  $ LowIncomeTracts     : num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
##  $ PovertyRate         : num [1:72864] 10 18.2 19.1 3.3 8.5 ...
##  $ MedianFamilyIncome  : num [1:72864] 74750 51875 52905 68079 77819 ...
##  $ LA1and10            : num [1:72864] 1 0 1 1 1 1 1 0 0 1 ...
##  $ LAhalfand10         : num [1:72864] 1 1 1 1 1 1 1 0 0 1 ...
##  $ LA1and20            : num [1:72864] 1 0 1 1 1 1 1 0 0 0 ...
##  $ LATracts_half       : num [1:72864] 1 1 1 1 1 1 1 0 0 0 ...
```

```
##  $ LATracts1          : num [1:72864] 1 0 1 1 1 1 1 0 0 0 ...
##  $ LATracts10         : num [1:72864] 0 0 0 0 0 0 0 0 0 1 ...
##  $ LATracts20         : num [1:72864] 0 0 0 0 0 0 0 0 0 0 ...
##  $ LATractsVehicle_20 : num [1:72864] 0 0 0 0 1 0 1 1 0 0 ...
##  $ LAPOP1_10          : num [1:72864] 1357 483 1418 1363 2643 ...
##  $ LAPOP05_10         : num [1:72864] 1732 1410 2765 3651 7778 ...
##  $ LAPOP1_20          : num [1:72864] 1357 483 1418 1363 2643 ...
##  $ LALOWI1_10         : num [1:72864] 322 145 697 410 623 ...
##  $ LALOWI05_10        : num [1:72864] 412 475 1350 1068 1913 ...
##  $ LALOWI1_20         : num [1:72864] 322 145 697 410 623 ...
##  $ lapophalf          : num [1:72864] 1732 1410 2765 3651 7778 ...
##  $ lapophalfshare     : num [1:72864] 0.906 0.65 0.82 0.832 0.722 ...
##  $ lalowihalf         : num [1:72864] 412 475 1350 1068 1913 ...
##  $ lalowihalfshare    : num [1:72864] 0.215 0.219 0.4 0.243 0.178 ...
##  $ lakidshalf         : num [1:72864] 466 448 745 847 2309 ...
##  $ lakidshalfshare    : num [1:72864] 0.244 0.207 0.221 0.193 0.215 ...
##  $ laseniorshalf      : num [1:72864] 199 139 346 767 840 ...
##  $ laseniorshalfshare : num [1:72864] 0.104 0.0642 0.1026 0.1748 0.078 ...
##  $ lawhitehalf        : num [1:72864] 1483 412 2115 3395 6299 ...
##  $ lawhitehalfshare   : num [1:72864] 0.776 0.19 0.627 0.774 0.585 ...
##  $ lablackhalf        : num [1:72864] 184 945 528 170 1001 ...
##  $ lablackhalfshare   : num [1:72864] 0.0964 0.4356 0.1566 0.0388 0.0929 ...
##  $ laasianhalf        : num [1:72864] 12.72 4 9.03 14.87 208.98 ...
##  $ laasianhalfshare   : num [1:72864] 0.00665 0.00184 0.00268 0.00339 0.01941 ...
##  $ lanhopihalf        : num [1:72864] 0 0 1 2.64 5.2 ...
##  $ lanhopihalfshare   : num [1:72864] 0 0 0.000296 0.000602 0.000483 ...
##  $ laaianhalf         : num [1:72864] 13.97 4.72 9.96 8.03 37.99 ...
##  $ laaianhalfshare    : num [1:72864] 0.00731 0.00217 0.00295 0.00183 0.00353 ...
##  $ laomultirhalf      : num [1:72864] 38.3 44 101.7 60.4 226.8 ...
##  $ laomultirhalfshare : num [1:72864] 0.0201 0.0203 0.0302 0.0138 0.0211 ...
##  $ lahisphalf         : num [1:72864] 39.7 34.6 76.4 61.4 276.9 ...
##  $ lahisphalfshare    : num [1:72864] 0.0208 0.0159 0.0227 0.014 0.0257 ...
##  $ lahunvhalf         : num [1:72864] 21.6 58.6 49.1 17.5 129.6 ...
##  $ lahunvhalfshare    : num [1:72864] 0.0311 0.0789 0.0391 0.0102 0.0317 ...
##  $ lasnaphalf         : num [1:72864] 101.9 127.4 100.2 67.7 339.1 ...
##  $ lasnaphalfshare    : num [1:72864] 0.147 0.1714 0.0798 0.0393 0.0831 ...
##  $ lapop1             : num [1:72864] 1357 483 1418 1363 2643 ...
##  $ lapop1share        : num [1:72864] 0.71 0.223 0.42 0.311 0.246 ...
##  $ lalowi1            : num [1:72864] 322 145 697 410 623 ...
##  $ lalowi1share       : num [1:72864] 0.1685 0.0669 0.2065 0.0934 0.0579 ...
##  $ lakids1            : num [1:72864] 364 175 377 346 715 ...
##  $ lakids1share       : num [1:72864] 0.1902 0.0805 0.1118 0.0789 0.0664 ...
##  $ laseniors1         : num [1:72864] 162 51 190 237 362 ...
##  $ laseniors1share    : num [1:72864] 0.085 0.0235 0.0563 0.0539 0.0336 ...
##  $ lawhite1           : num [1:72864] 1162 128 1168 1233 2168 ...
##  $ lawhite1share      : num [1:72864] 0.6076 0.0591 0.3463 0.2812 0.2014 ...
##  $ lablack1           : num [1:72864] 147.5 335.4 202.5 80.9 343.2 ...
##  $ lablack1share      : num [1:72864] 0.0771 0.1546 0.06 0.0185 0.0319 ...
##  $ laasian1           : num [1:72864] 11.02 1.59 7.82 6.88 47.49 ...
##  $ laasian1share      : num [1:72864] 0.005761 0.000733 0.002319 0.001567 0.004412 ...
##  $ lanhopi1           : num [1:72864] 0 0 0 2 0.953 ...
##  $ lanhopi1share      : num [1:72864] 0.00 0.00 0.00 4.56e-04 8.85e-05 ...
##  $ laaian1            : num [1:72864] 9.997 0.264 1.656 3.54 13.561 ...
##  $ laaian1share       : num [1:72864] 0.005228 0.000121 0.000491 0.000807 0.00126 ...
```

```
##  $ laomultir1         : num [1:72864] 27.2 18 37.9 36.8 70.1 ...
##  $ laomultir1share    : num [1:72864] 0.01421 0.00828 0.01123 0.0084 0.00651 ...
##  $ lahisp1            : num [1:72864] 29.7 11.2 31.2 29.9 85.8 ...
##  $ lahisp1share       : num [1:72864] 0.01554 0.00515 0.00926 0.00681 0.00797 ...
##  $ lahunv1            : num [1:72864] 9.77 21.64 13.31 8.78 44.66 ...
##  $ lahunv1share       : num [1:72864] 0.0141 0.0291 0.0106 0.0051 0.0109 ...
##  $ lasnap1            : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
##  $ lasnap1share       : num [1:72864] 0.1148 0.0561 0.04 0.0142 0.0294 ...
##  $ lapop10            : num [1:72864] 0 0 0 0 0 ...
##  $ lapop10share       : num [1:72864] 0 0 0 0 0 ...
##  $ lalowi10           : num [1:72864] 0 0 0 0 0 ...
##  $ lalowi10share      : num [1:72864] 0 0 0 0 0 ...
##  $ lakids10           : num [1:72864] 0 0 0 0 0 ...
##  $ lakids10share      : num [1:72864] 0 0 0 0 0 ...
##  $ laseniors10        : num [1:72864] 0 0 0 0 0 ...
##  $ laseniors10share   : num [1:72864] 0 0 0 0 0 ...
##  $ lawhite10          : num [1:72864] 0 0 0 0 0 ...
##  $ lawhite10share     : num [1:72864] 0 0 0 0 0 ...
##  $ lablack10          : num [1:72864] 0 0 0 0 0 ...
##  $ lablack10share     : num [1:72864] 0 0 0 0 0 ...
##  $ laasian10          : num [1:72864] 0 0 0 0 0 ...
##  $ laasian10share     : num [1:72864] 0 0 0 0 0 ...
##  $ lanhopi10          : num [1:72864] 0 0 0 0 0 ...
##  $ lanhopi10share     : num [1:72864] 0 0 0 0 0 ...
##   [list output truncated]
```

```r
str(DATA2019)
```

```
## tibble [72,531 x 147] (S3: tbl_df/tbl/data.frame)
##  $ CensusTract        : chr [1:72531] "01001020100" "01001020200" "01001020300" "01001020400" ...
##  $ State              : chr [1:72531] "Alabama" "Alabama" "Alabama" "Alabama" ...
##  $ County             : chr [1:72531] "Autauga County" "Autauga County" "Autauga County" "Autauga Co
##  $ Urban              : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
##  $ Pop2010            : num [1:72531] 1912 2170 3373 4386 10766 ...
##  $ OHU2010            : num [1:72531] 693 743 1256 1722 4082 ...
##  $ GroupQuartersFlag  : num [1:72531] 0 0 0 0 0 0 0 0 0 0 ...
##  $ NUMGQTRS           : num [1:72531] 0 181 0 0 181 0 36 0 0 14 ...
##  $ PCTGQTRS           : num [1:72531] 0 8.34 0 0 1.68 ...
##  $ LILATracts_1And10   : num [1:72531] 0 1 0 0 0 1 1 0 0 0 ...
##  $ LILATracts_halfAnd10: num [1:72531] 0 1 0 0 0 1 1 0 0 0 ...
##  $ LILATracts_1And20   : num [1:72531] 0 1 0 0 0 1 1 0 0 0 ...
##  $ LILATracts_Vehicle  : num [1:72531] 0 0 0 0 0 0 0 0 0 0 ...
##  $ HUNVFlag           : num [1:72531] 0 0 0 0 1 0 0 0 1 0 ...
##  $ LowIncomeTracts    : num [1:72531] 0 1 0 0 0 1 1 0 0 0 ...
##  $ PovertyRate        : num [1:72531] 11.34 17.88 15.05 2.85 15.15 ...
##  $ MedianFamilyIncome  : num [1:72531] 81250 49000 62609 70607 96334 ...
##  $ LA1and10           : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
##  $ LAhalfand10        : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
##  $ LA1and20           : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
##  $ LATracts_half      : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
##  $ LATracts1          : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
##  $ LATracts10         : num [1:72531] 0 0 0 0 0 0 0 0 0 0 ...
##  $ LATracts20         : num [1:72531] 0 0 0 0 0 0 0 0 0 0 ...
##  $ LATractsVehicle_20  : num [1:72531] 0 0 0 0 1 0 0 0 1 0 ...
##  $ LAPOP1_10          : num [1:72531] 1896 1261 1552 1363 2643 ...
```

```
##  $ LAPOP05_10          : num [1:72531] 1912 2170 2857 3651 7778 ...
##  $ LAPOP1_20           : num [1:72531] 1896 1261 1552 1363 2643 ...
##  $ LALOWI1_10          : num [1:72531] 461 604 478 343 586 ...
##  $ LALOWI05_10         : num [1:72531] 467 962 971 893 1719 ...
##  $ LALOWI1_20          : num [1:72531] 461 604 478 343 586 ...
##  $ lapophalf           : num [1:72531] 1912 2170 2857 3651 7778 ...
##  $ lapophalfshare      : num [1:72531] 100 100 84.7 83.2 72.2 ...
##  $ lalowihalf          : num [1:72531] 467 962 971 893 1719 ...
##  $ lalowihalfshare     : num [1:72531] 24.4 44.3 28.8 20.4 16 ...
##  $ lakidshalf          : num [1:72531] 507 606 771 847 2309 ...
##  $ lakidshalfshare     : num [1:72531] 26.5 27.9 22.9 19.3 21.5 ...
##  $ laseniorshalf       : num [1:72531] 221 214 358 767 840 ...
##  $ laseniorshalfshare  : num [1:72531] 11.56 9.86 10.6 17.48 7.8 ...
##  $ lawhitehalf         : num [1:72531] 1622 888 2177 3395 6299 ...
##  $ lawhitehalfshare    : num [1:72531] 84.8 40.9 64.5 77.4 58.5 ...
##  $ lablackhalf         : num [1:72531] 217 1217 554 170 1001 ...
##  $ lablackhalfshare    : num [1:72531] 11.35 56.08 16.43 3.88 9.29 ...
##  $ laasianhalf         : num [1:72531] 14 5 10.3 14.9 209 ...
##  $ laasianhalfshare    : num [1:72531] 0.732 0.23 0.304 0.339 1.941 ...
##  $ lanhopihalf         : num [1:72531] 0 0 1 2.64 5.2 ...
##  $ lanhopihalfshare    : num [1:72531] 0 0 0.0296 0.0602 0.0483 ...
##  $ laaianhalf          : num [1:72531] 14 5 10.24 8.03 37.99 ...
##  $ laaianhalfshare     : num [1:72531] 0.732 0.23 0.304 0.183 0.353 ...
##  $ laomultirhalf       : num [1:72531] 45 55 104.5 60.4 226.8 ...
##  $ laomultirhalfshare  : num [1:72531] 2.35 2.53 3.1 1.38 2.11 ...
##  $ lahisphalf          : num [1:72531] 44 75 77.6 61.4 276.9 ...
##  $ lahisphalfshare     : num [1:72531] 2.3 3.46 2.3 1.4 2.57 ...
##  $ lahunvhalf          : num [1:72531] 5.49 92.67 38.77 19.42 163.89 ...
##  $ lahunvhalfshare     : num [1:72531] 0.792 12.473 3.087 1.128 4.015 ...
##  $ lasnaphalf          : num [1:72531] 92.4 161.2 138.8 84.1 235 ...
##  $ lasnaphalfshare     : num [1:72531] 13.33 21.7 11.05 4.88 5.76 ...
##  $ lapop1              : num [1:72531] 1896 1261 1552 1363 2643 ...
##  $ lapop1share         : num [1:72531] 99.2 58.1 46 31.1 24.6 ...
##  $ lalowi1             : num [1:72531] 461 604 478 343 586 ...
##  $ lalowi1share        : num [1:72531] 24.11 27.83 14.18 7.83 5.45 ...
##  $ lakids1             : num [1:72531] 504 406 416 346 715 ...
##  $ lakids1share        : num [1:72531] 26.33 18.69 12.34 7.89 6.64 ...
##  $ laseniors1          : num [1:72531] 219 127 201 237 362 ...
##  $ laseniors1share     : num [1:72531] 11.44 5.83 5.96 5.39 3.36 ...
##  $ lawhite1            : num [1:72531] 1611 357 1242 1233 2168 ...
##  $ lawhite1share       : num [1:72531] 84.3 16.4 36.8 28.1 20.1 ...
##  $ lablack1            : num [1:72531] 213.6 854.2 255.1 80.9 343.2 ...
##  $ lablack1share       : num [1:72531] 11.17 39.36 7.56 1.85 3.19 ...
##  $ laasian1            : num [1:72531] 13.8 4 8.03 6.88 47.49 ...
##  $ laasian1share       : num [1:72531] 0.722 0.184 0.238 0.157 0.441 ...
##  $ lanhopi1            : num [1:72531] 0 0 0 2 0.953 ...
##  $ lanhopi1share       : num [1:72531] 0 0 0 0.0456 0.00885 ...
##  $ laaian1             : num [1:72531] 14 4.4 2.08 3.54 13.56 ...
##  $ laaian1share        : num [1:72531] 0.7322 0.2026 0.0615 0.0807 0.126 ...
##  $ laomultir1          : num [1:72531] 44.1 41.8 44.8 36.8 70.1 ...
##  $ laomultir1share     : num [1:72531] 2.307 1.926 1.328 0.84 0.651 ...
##  $ lahisp1             : num [1:72531] 43.4 33 36.3 29.9 85.8 ...
##  $ lahisp1share        : num [1:72531] 2.268 1.52 1.075 0.681 0.797 ...
##  $ lahunv1             : num [1:72531] 5.49 66.9 0 7.94 55.21 ...
```

```
##  $ lahunv1share        : num [1:72531] 0.792 9.005 0 0.461 1.353 ...
##  $ lasnap1             : num [1:72531] 91.6 96.2 73.7 30.3 83.1 ...
##  $ lasnap1share        : num [1:72531] 13.22 12.95 5.87 1.76 2.04 ...
##  $ lapop10             : num [1:72531] NA NA NA NA NA ...
##  $ lapop10share        : num [1:72531] NA NA NA NA NA ...
##  $ lalowi10            : num [1:72531] NA NA NA NA NA ...
##  $ lalowi10share       : num [1:72531] NA NA NA NA NA ...
##  $ lakids10            : num [1:72531] NA NA NA NA NA ...
##  $ lakids10share       : num [1:72531] NA NA NA NA NA ...
##  $ laseniors10         : num [1:72531] NA NA NA NA NA ...
##  $ laseniors10share    : num [1:72531] NA NA NA NA NA ...
##  $ lawhite10           : num [1:72531] NA NA NA NA NA ...
##  $ lawhite10share      : num [1:72531] NA NA NA NA NA ...
##  $ lablack10           : num [1:72531] NA NA NA NA NA ...
##  $ lablack10share      : num [1:72531] NA NA NA NA NA ...
##  $ laasian10           : num [1:72531] NA NA NA NA NA NA NA 0 0 0 ...
##  $ laasian10share      : num [1:72531] NA NA NA NA NA NA NA 0 0 0 ...
##  $ lanhopi10           : num [1:72531] NA NA NA NA NA NA NA 0 0 0 ...
##  $ lanhopi10share      : num [1:72531] NA NA NA NA NA NA NA 0 0 0 ...
##   [list output truncated]
```

```r
# To Unify variable names
names(DATA2015) <- tolower(names(DATA2015))
names(DATA2019) <- tolower(names(DATA2019))
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(knitr)
library(ggplot2)
```

```r
# Select variables we need
data2015_clean <- DATA2015 %>%
  select(
    censustract,
    state,
    county,
    urban,
    pop2010,
    ohu2010,
    laseniors1,
    laseniors10,
    lasnap1,
    laseniors1share,
    laseniors10share,
    lasnap1share,
```

```r
    lasnap10share,
    lasnap10,
    laseniors20,
    laseniorshalf,
    laseniorshalfshare,
    tractseniors,
    tractsnap,
  )

data2019_clean <- DATA2019 %>%
  select(
    censustract,
    state,
    county,
    urban,
    pop2010,
    ohu2010,
    laseniors1,
    laseniors10,
    laseniors1share,
    laseniors10share,
    lasnap1share,
    lasnap10share,
    lasnap1,
    lasnap10,
    laseniors20,
    laseniorshalf,
    laseniorshalfshare,
    tractseniors,
    tractsnap,
)
```

```r
numeric_vars <- c(
  "urban",
  "pop2010",
  "ohu2010",
  "laseniors1",
  "laseniors10",
  "laseniors1share",
  "laseniors10share",
  "lasnap1",
  "lasnap10",
  "lasnap1share",
  "lasnap10share",
  "tractseniors",
  "tractsnap"
)

data2015_clean <- data2015_clean %>%
  mutate(
    across(
      any_of(numeric_vars),
      as.numeric
    )
```

```r
  )

data2019_clean <- data2019_clean %>%
  mutate(
    across(
      any_of(numeric_vars),
      as.numeric
    )
  )

combined_data <- bind_rows(
  data2015_clean,
  data2019_clean
)

combined_data
```

```
## # A tibble: 145,395 x 19
##    censustract state county urban pop2010 ohu2010 laseniors1 laseniors10 lasnap1
##    <chr>       <chr> <chr>  <dbl>   <dbl>   <dbl>      <dbl>       <dbl>   <dbl>
##  1 01001020100 Alab~ Autau~     1    1912     693      162.          0      79.5
##  2 01001020200 Alab~ Autau~     1    2170     743       51.0         0      41.7
##  3 01001020300 Alab~ Autau~     1    3373    1256      190.          0      50.3
##  4 01001020400 Alab~ Autau~     1    4386    1722      237.          0      24.4
##  5 01001020500 Alab~ Autau~     1   10766    4082      362.          0     120.
##  6 01001020600 Alab~ Autau~     1    3668    1311      264.          0      87.5
##  7 01001020700 Alab~ Autau~     1    2891    1188      109.          0     110.
##  8 01001020801 Alab~ Autau~     0    3081    1074      341.          0     105.
##  9 01001020802 Alab~ Autau~     0   10435    3694     1060.         24.4   365.
## 10 01001020900 Alab~ Autau~     0    5675    2067      630.        242.    391.
## # i 145,385 more rows
## # i 10 more variables: laseniors1share <dbl>, laseniors10share <dbl>,
## #   lasnap1share <dbl>, lasnap10share <dbl>, lasnap10 <dbl>, laseniors20 <dbl>,
## #   laseniorshalf <dbl>, laseniorshalfshare <dbl>, tractseniors <dbl>,
## #   tractsnap <dbl>
```

```r
sapply(
  data2015_clean[numeric_vars],
  class
)
```

```
##            urban          pop2010          ohu2010       laseniors1
##        "numeric"        "numeric"        "numeric"        "numeric"
##      laseniors10  laseniors1share laseniors10share          lasnap1
##        "numeric"        "numeric"        "numeric"        "numeric"
##         lasnap10     lasnap1share    lasnap10share     tractseniors
##        "numeric"        "numeric"        "numeric"        "numeric"
##        tractsnap
##        "numeric"
```

```r
sapply(
  data2019_clean[numeric_vars],
  class
)
```

```
##           urban          pop2010          ohu2010       laseniors1
##       "numeric"        "numeric"        "numeric"        "numeric"
##     laseniors10 laseniors1share laseniors10share          lasnap1
##       "numeric"        "numeric"        "numeric"        "numeric"
##        lasnap10     lasnap1share     lasnap10share    tractseniors
##       "numeric"        "numeric"        "numeric"        "numeric"
##       tractsnap
##       "numeric"
```

```
data2015_clean <- data2015_clean %>%
  mutate(year = 2015)

data2019_clean <- data2019_clean %>%
  mutate(year = 2019)
```

```
#Create Urban/Rural
data2015_clean <- data2015_clean %>%
  mutate(
    year = 2015,
    area = if_else(
      urban == 1,
      "Urban",
      "Rural",
      missing = NA_character_
    )
  )

data2019_clean <- data2019_clean %>%
  mutate(
    year = 2019,
    area = if_else(
      urban == 1,
      "Urban",
      "Rural",
      missing = NA_character_
    )
  )
```

```
# 2015 of senior and snap
data2015_clean <- data2015_clean %>%
  mutate(
    senior_la_n = case_when(
      urban == 1 ~ laseniors1,
      urban == 0 ~ laseniors10,
      TRUE ~ NA_real_
    ),

    snap_la_n = case_when(
      urban == 1 ~ lasnap1,
      urban == 0 ~ lasnap10,
      TRUE ~ NA_real_
    )
  )

# 2019 of senior and snap
```

```r
data2019_clean <- data2019_clean %>%
  mutate(
    senior_la_n = case_when(
      urban == 1 ~ laseniors1,
      urban == 0 ~ laseniors10,
      TRUE ~ NA_real_
    ),

    snap_la_n = case_when(
      urban == 1 ~ lasnap1,
      urban == 0 ~ lasnap10,
      TRUE ~ NA_real_
    )
  )
```

```r
# Show precentage
# 2015
data2015_clean <- data2015_clean %>%
  mutate(
    senior_la_pct = case_when(
      urban == 1 ~ laseniors1share * 100,
      urban == 0 ~ laseniors10share * 100,
      TRUE ~ NA_real_
    ),

    snap_la_pct = case_when(
      urban == 1 ~ lasnap1share * 100,
      urban == 0 ~ lasnap10share * 100,
      TRUE ~ NA_real_
    )
  )

# 2019
data2019_clean <- data2019_clean %>%
  mutate(
    senior_la_pct = case_when(
      urban == 1 ~ laseniors1share,
      urban == 0 ~ laseniors10share,
      TRUE ~ NA_real_
    ),

    snap_la_pct = case_when(
      urban == 1 ~ lasnap1share,
      urban == 0 ~ lasnap10share,
      TRUE ~ NA_real_
    )
  )
```

```r
# Seniors
data2015_clean <- data2015_clean %>%
  mutate(
    senior_la_n = if_else(
      urban == 1,
      laseniors1,
```

```r
      laseniors10
    )
  )

# SNAP
data2015_clean <- data2015_clean %>%
  mutate(
    snap_la_n = if_else(
      urban == 1,
      lasnap1,
      lasnap10
    )
  )

# Seniors
data2019_clean <- data2019_clean %>%
  mutate(
    senior_la_n = if_else(
      urban == 1,
      laseniors1,
      laseniors10
    )
  )

# SNAP
data2019_clean <- data2019_clean %>%
  mutate(
    snap_la_n = if_else(
      urban == 1,
      lasnap1,
      lasnap10
    )
  )
```

```r
summary(data2015_clean)
```

```
##  censustract           state              county              urban       
##  Length:72864       Length:72864       Length:72864       Min.   :0.0000  
##  Class :character   Class :character   Class :character   1st Qu.:1.0000  
##  Mode  :character   Mode  :character   Mode  :character   Median :1.0000  
##                                                           Mean   :0.7572  
##                                                           3rd Qu.:1.0000  
##                                                           Max.   :1.0000  
##     pop2010         ohu2010        laseniors1       laseniors10    
##  Min.   :    0   Min.   :    0   Min.   :    0.0   Min.   :   0.0  
##  1st Qu.: 2884   1st Qu.: 1102   1st Qu.:    0.0   1st Qu.:   0.0  
##  Median : 4002   Median : 1521   Median :  111.2   Median :   0.0  
##  Mean   : 4237   Mean   : 1602   Mean   :  230.6   Mean   :  12.1  
##  3rd Qu.: 5323   3rd Qu.: 2018   3rd Qu.:  390.6   3rd Qu.:   0.0  
##  Max.   :37452   Max.   :16043   Max.   :10225.7   Max.   :2531.0  
##     lasnap1         laseniors1share   laseniors10share   lasnap1share    
##  Min.   :   0.00   Min.   :0.00000   Min.   :0.00000   Min.   :0.00000  
##  1st Qu.:   0.00   1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.00000  
##  Median :  20.11   Median :0.02761   Median :0.00000   Median :0.01323
```

```
## Mean   :  71.48   Mean   :0.05636   Mean   :0.00413   Mean   :0.04602
## 3rd Qu.: 103.09   3rd Qu.:0.09887   3rd Qu.:0.00000   3rd Qu.:0.06732
## Max.   :1521.24   Max.   :1.00000   Max.   :0.57680   Max.   :1.00000
## lasnap10share        lasnap10        laseniors20       laseniorshalf
## Min.   :0.000000   Min.   :   0.00   Min.   :   0.000   Min.   :    0.0
## 1st Qu.:0.000000   1st Qu.:   0.00   1st Qu.:   0.000   1st Qu.:  153.7
## Median :0.000000   Median :   0.00   Median :   0.000   Median :  346.0
## Mean   :0.003116   Mean   :   3.71   Mean   :   1.492   Mean   :  395.2
## 3rd Qu.:0.000000   3rd Qu.:   0.00   3rd Qu.:   0.000   3rd Qu.:  567.2
## Max.   :0.888889   Max.   :1108.69   Max.   :2081.129   Max.   :15260.1
## laseniorshalfshare  tractseniors       tractsnap          year
## Min.   :0.00000    Min.   :    0.0   Min.   :   0   Min.   :2015
## 1st Qu.:0.04248    1st Qu.:  318.0   1st Qu.:  69   1st Qu.:2015
## Median :0.09145    Median :  495.0   Median : 158   Median :2015
## Mean   :0.09642    Mean   :  552.6   Mean   : 207   Mean   :2015
## 3rd Qu.:0.13747    3rd Qu.:  716.0   3rd Qu.: 293   3rd Qu.:2015
## Max.   :1.00000    Max.   :17271.0   Max.   :2152   Max.   :2015
##     area             senior_la_n          snap_la_n          senior_la_pct
## Length:72864       Min.   :    0.000   Min.   :   0.0000   Min.   :  0.0000
## Class :character   1st Qu.:    0.000   1st Qu.:   0.0000   1st Qu.:  0.0000
## Mode  :character   Median :    5.792   Median :   0.9336   Median :  0.1582
##                    Mean   :  123.775   Mean   :  38.2526   Mean   :  2.9614
##                    3rd Qu.:  167.168   3rd Qu.:  37.7681   3rd Qu.:  4.1252
##                    Max.   :10225.695   Max.   :1521.2415   Max.   :100.0000
##  snap_la_pct
## Min.   :  0.00000
## 1st Qu.:  0.00000
## Median :  0.06551
## Mean   :  2.48353
## 3rd Qu.:  2.41834
## Max.   :100.00000
```

```
summary(data2019_clean)
```

```
## censustract           state              county              urban
## Length:72531       Length:72531       Length:72531       Min.   :0.0000
## Class :character   Class :character   Class :character   1st Qu.:1.0000
## Mode  :character   Mode  :character   Mode  :character   Median :1.0000
##                                                          Mean   :0.7606
##                                                          3rd Qu.:1.0000
##                                                          Max.   :1.0000
##
##    pop2010         ohu2010        laseniors1        laseniors10
## Min.   :    1   Min.   :    0   Min.   :    0.00   Min.   :   0.00
## 1st Qu.: 2899   1st Qu.: 1108   1st Qu.:   73.79   1st Qu.:   7.58
## Median : 4011   Median : 1525   Median :  259.28   Median :  51.70
## Mean   : 4257   Mean   : 1609   Mean   :  319.30   Mean   : 111.16
## 3rd Qu.: 5330   3rd Qu.: 2021   3rd Qu.:  484.00   3rd Qu.: 162.38
## Max.   :37452   Max.   :16043   Max.   :10348.58   Max.   :2531.00
##                                 NA's   :19989      NA's   :64765
## laseniors1share   laseniors10share  lasnap1share     lasnap10share
## Min.   :  0.000   Min.   : 0.00    Min.   :  0.000   Min.   : 0.00
## 1st Qu.:  1.879   1st Qu.: 0.20    1st Qu.:  0.794   1st Qu.: 0.11
## Median :  6.549   Median : 1.50    Median :  3.431   Median : 0.85
## Mean   :  7.826   Mean   : 3.81    Mean   :  5.996   Mean   : 2.67
```

```
##  3rd Qu.: 12.227   3rd Qu.: 5.30    3rd Qu.:  8.660   3rd Qu.: 3.01
##  Max.   :100.000   Max.   :57.68    Max.   :100.000   Max.   :84.53
##  NA's   :19989     NA's   :64765    NA's   :19966     NA's   :64666
##     lasnap1           lasnap10        laseniors20       laseniorshalf
##  Min.   :   0.00   Min.   :  0.00   Min.   :   0.00   Min.   :    0.0
##  1st Qu.:  11.89   1st Qu.:  1.77   1st Qu.:   0.42   1st Qu.:  195.0
##  Median :  52.93   Median : 11.33   Median :  12.40   Median :  372.7
##  Mean   :  92.52   Mean   : 31.93   Mean   :  63.29   Mean   :  422.9
##  3rd Qu.: 133.53   3rd Qu.: 37.97   3rd Qu.:  72.97   3rd Qu.:  585.8
##  Max.   :1582.31   Max.   :995.18   Max.   :2081.13   Max.   :15261.1
##  NA's   :19989     NA's   :64765    NA's   :71025     NA's   :4568
##  laseniorshalfshare  tractseniors       tractsnap           year
##  Min.   :  0.000    Min.   :    0.0   Min.   :   0.0   Min.   :2019
##  1st Qu.:  5.286    1st Qu.:  320.0   1st Qu.:  67.0   1st Qu.:2019
##  Median :  9.768    Median :  497.0   Median : 152.0   Median :2019
##  Mean   : 10.326    Mean   :  555.2   Mean   : 201.8   Mean   :2019
##  3rd Qu.: 14.080    3rd Qu.:  718.0   3rd Qu.: 282.0   3rd Qu.:2019
##  Max.   :100.000    Max.   :17271.0   Max.   :2175.0   Max.   :2019
##  NA's   :4568       NA's   :4         NA's   :4
##      area            senior_la_n        snap_la_n         senior_la_pct
##  Length:72531       Min.   :    0.0   Min.   :   0.000   Min.   :  0.000
##  Class :character   1st Qu.:   25.8   1st Qu.:   4.591   1st Qu.:  0.668
##  Mode  :character   Median :  126.0   Median :  24.812   Median :  3.157
##                     Mean   :  211.4   Mean   :  61.941   Mean   :  5.086
##                     3rd Qu.:  308.0   3rd Qu.:  78.624   3rd Qu.:  7.530
##                     Max.   :10348.6   Max.   :1582.311   Max.   :100.000
##                     NA's   :29953     NA's   :29953      NA's   :29953
##   snap_la_pct
##  Min.   : 0.000
##  1st Qu.: 0.304
##  Median : 1.600
##  Mean   : 4.062
##  3rd Qu.: 4.869
##  Max.   :85.091
##  NA's   :29918
```

```r
#data2019_clean <- data2019_clean %>%
#  mutate(
#    across(
#      c(laseniors1, laseniors10, lasnap1, lasnap10,
#        laseniors20,laseniorshalf,laseniorshalfshare,senior_la_n, snap_la_n),
#      ~ as.numeric(ifelse(.x == "NULL", "0", .x))
#    )
#  )
```

```r
summary(data2019_clean)
```

```
##  censustract           state              county              urban
##  Length:72531       Length:72531       Length:72531       Min.   :0.0000
##  Class :character   Class :character   Class :character   1st Qu.:1.0000
##  Mode  :character   Mode  :character   Mode  :character   Median :1.0000
##                                                           Mean   :0.7606
##                                                           3rd Qu.:1.0000
##                                                           Max.   :1.0000
##
```

```
##     pop2010          ohu2010         laseniors1         laseniors10
##  Min.   :     1  Min.   :     0  Min.   :    0.00   Min.   :   0.00
##  1st Qu.: 2899   1st Qu.: 1108   1st Qu.:   73.79   1st Qu.:   7.58
##  Median : 4011   Median : 1525   Median :  259.28   Median :  51.70
##  Mean   : 4257   Mean   : 1609   Mean   :  319.30   Mean   : 111.16
##  3rd Qu.: 5330   3rd Qu.: 2021   3rd Qu.:  484.00   3rd Qu.: 162.38
##  Max.   :37452   Max.   :16043   Max.   :10348.58   Max.   :2531.00
##                                  NA's   :19989      NA's   :64765
##  laseniors1share   laseniors10share  lasnap1share     lasnap10share
##  Min.   :  0.000   Min.   : 0.00    Min.   :  0.000   Min.   : 0.00
##  1st Qu.:  1.879   1st Qu.: 0.20    1st Qu.:  0.794   1st Qu.: 0.11
##  Median :  6.549   Median : 1.50    Median :  3.431   Median : 0.85
##  Mean   :  7.826   Mean   : 3.81    Mean   :  5.996   Mean   : 2.67
##  3rd Qu.: 12.227   3rd Qu.: 5.30    3rd Qu.:  8.660   3rd Qu.: 3.01
##  Max.   :100.000   Max.   :57.68    Max.   :100.000   Max.   :84.53
##  NA's   :19989     NA's   :64765    NA's   :19966     NA's   :64666
##     lasnap1            lasnap10       laseniors20       laseniorshalf
##  Min.   :   0.00   Min.   :  0.00   Min.   :   0.00   Min.   :    0.0
##  1st Qu.:  11.89   1st Qu.:  1.77   1st Qu.:   0.42   1st Qu.:  195.0
##  Median :  52.93   Median : 11.33   Median :  12.40   Median :  372.7
##  Mean   :  92.52   Mean   : 31.93   Mean   :  63.29   Mean   :  422.9
##  3rd Qu.: 133.53   3rd Qu.: 37.97   3rd Qu.:  72.97   3rd Qu.:  585.8
##  Max.   :1582.31   Max.   :995.18   Max.   :2081.13   Max.   :15261.1
##  NA's   :19989     NA's   :64765    NA's   :71025     NA's   :4568
##  laseniorshalfshare  tractseniors       tractsnap           year
##  Min.   :  0.000    Min.   :    0.0   Min.   :   0.0   Min.   :2019
##  1st Qu.:  5.286    1st Qu.:  320.0   1st Qu.:  67.0   1st Qu.:2019
##  Median :  9.768    Median :  497.0   Median : 152.0   Median :2019
##  Mean   : 10.326    Mean   :  555.2   Mean   : 201.8   Mean   :2019
##  3rd Qu.: 14.080    3rd Qu.:  718.0   3rd Qu.: 282.0   3rd Qu.:2019
##  Max.   :100.000    Max.   :17271.0   Max.   :2175.0   Max.   :2019
##  NA's   :4568       NA's   :4         NA's   :4
##      area            senior_la_n        snap_la_n        senior_la_pct
##  Length:72531       Min.   :    0.0   Min.   :   0.000   Min.   :  0.000
##  Class :character   1st Qu.:   25.8   1st Qu.:   4.591   1st Qu.:  0.668
##  Mode  :character   Median :  126.0   Median :  24.812   Median :  3.157
##                     Mean   :  211.4   Mean   :  61.941   Mean   :  5.086
##                     3rd Qu.:  308.0   3rd Qu.:  78.624   3rd Qu.:  7.530
##                     Max.   :10348.6   Max.   :1582.311   Max.   :100.000
##                     NA's   :29953     NA's   :29953      NA's   :29953
##   snap_la_pct
##  Min.   : 0.000
##  1st Qu.: 0.304
##  Median : 1.600
##  Mean   : 4.062
##  3rd Qu.: 4.869
##  Max.   :85.091
##  NA's   :29918
```

```r
summary_stats <- bind_rows(

  data2015_clean %>%
    transmute(
      year = 2015,
```

```r
      Senior = senior_la_pct,
      SNAP = snap_la_pct
    ),

  data2019_clean %>%
    transmute(
      year = 2019,
      Senior = senior_la_pct,
      SNAP = snap_la_pct
    )

) %>%

  pivot_longer(
    cols = c(Senior, SNAP),
    names_to = "Group",
    values_to = "Low_Access_Pct"
  ) %>%

  group_by(year, Group) %>%

  summarise(
    Min = min(Low_Access_Pct, na.rm = TRUE),
    Q1 = quantile(Low_Access_Pct, 0.25, na.rm = TRUE),
    Median = median(Low_Access_Pct, na.rm = TRUE),
    Mean = mean(Low_Access_Pct, na.rm = TRUE),
    Q3 = quantile(Low_Access_Pct, 0.75, na.rm = TRUE),
    Max = max(Low_Access_Pct, na.rm = TRUE),
    Missing = sum(is.na(Low_Access_Pct)),
    .groups = "drop"
  ) %>%

  mutate(
    across(
      c(Min, Q1, Median, Mean, Q3, Max),
      ~ round(.x, 2)
    )
  )

summary_stats
```

```
## # A tibble: 4 x 9
##    year Group    Min    Q1 Median  Mean    Q3   Max Missing
##   <dbl> <chr>  <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl>   <int>
## 1  2015 SNAP       0  0      0.07  2.48  2.42 100         0
## 2  2015 Senior     0  0      0.16  2.96  4.13 100         0
## 3  2019 SNAP       0  0.3    1.6   4.06  4.87  85.1   29918
## 4  2019 Senior     0  0.67   3.16  5.09  7.53 100     29953
```

```r
kable(
  summary_stats,
  caption = "Summary Statistics for Senior and SNAP Low-Access Shares, 2015 and 2019",
  col.names = c(
    "Year",
```

```
    "Group",
    "Min (%)",
    "Q1 (%)",
    "Median (%)",
    "Mean (%)",
    "Q3 (%)",
    "Max (%)",
    "Missing"
  ),
  align = c(
    "c", "l",
    "r", "r", "r", "r", "r", "r", "r"
  )
)
```

Table 1: Summary Statistics for Senior and SNAP Low-Access Shares, 2015 and 2019

Year	Group	Min (%)	Q1 (%)	Median (%)	Mean (%)	Q3 (%)	Max (%)	Missing
2015	SNAP	0	0.00	0.07	2.48	2.42	100.00	0
2015	Senior	0	0.00	0.16	2.96	4.13	100.00	0
2019	SNAP	0	0.30	1.60	4.06	4.87	85.09	29918
2019	Senior	0	0.67	3.16	5.09	7.53	100.00	29953

```
# Check whether over 100
data2015_clean %>%
  summarise(
    senior_over_100 = sum(senior_la_pct > 100, na.rm = TRUE),
    snap_over_100 = sum(snap_la_pct > 100, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 2
##   senior_over_100 snap_over_100
##             <int>         <int>
## 1               0             1
```

```
data2019_clean %>%
  summarise(
    senior_over_100 = sum(senior_la_pct > 100, na.rm = TRUE),
    snap_over_100 = sum(snap_la_pct > 100, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 2
##   senior_over_100 snap_over_100
##             <int>         <int>
## 1               0             0
```

```
data2015_clean %>%
  filter(snap_la_pct > 100) %>%
  select(
    censustract,
    urban,
    area,
    lasnap1share,
```

```
    lasnap10share,
    snap_la_pct
  )
```

```
## # A tibble: 1 x 6
##   censustract urban area  lasnap1share lasnap10share snap_la_pct
##   <chr>       <dbl> <chr>        <dbl>         <dbl>       <dbl>
## 1 22079980000     1 Urban         1.00             0        100.
```

```
data2019_clean %>%
  filter(snap_la_pct > 100) %>%
  select(
    censustract,
    urban,
    area,
    lasnap1share,
    lasnap10share,
    snap_la_pct
  )
```

```
## # A tibble: 0 x 6
## # i 6 variables: censustract <chr>, urban <dbl>, area <chr>,
## #   lasnap1share <dbl>, lasnap10share <dbl>, snap_la_pct <dbl>
```

```
options(digits = 15)

data2015_clean %>%
  filter(snap_la_pct > 100) %>%
  select(
    censustract,
    urban,
    area,
    lasnap1share,
    lasnap10share,
    snap_la_pct
  )
```

```
## # A tibble: 1 x 6
##   censustract urban area  lasnap1share lasnap10share snap_la_pct
##   <chr>       <dbl> <chr>        <dbl>         <dbl>       <dbl>
## 1 22079980000     1 Urban         1.00             0        100.
```

```
data2019_clean %>%
  filter(snap_la_pct > 100) %>%
  select(
    censustract,
    urban,
    area,
    lasnap1share,
    lasnap10share,
    snap_la_pct
  )
```

```
## # A tibble: 0 x 6
## # i 6 variables: censustract <chr>, urban <dbl>, area <chr>,
## #   lasnap1share <dbl>, lasnap10share <dbl>, snap_la_pct <dbl>
```

```r
# Show precentage
#data2015_clean <- data2015_clean %>%
#  mutate(
#    senior_la_pct = senior_la_rate * 100,
#    snap_la_pct = snap_la_rate * 100
#  )

#data2019_clean <- data2019_clean %>%
#  mutate(
#    senior_la_pct = senior_la_rate * 100,
#    snap_la_pct = snap_la_rate * 100
#  )
```

```r
combined_data <- bind_rows(
  data2015_clean,
  data2019_clean
)
combined_data
```

```
## # A tibble: 145,395 x 25
##    censustract state county urban pop2010 ohu2010 laseniors1 laseniors10 lasnap1
##    <chr>       <chr> <chr>  <dbl>   <dbl>   <dbl>      <dbl>       <dbl>   <dbl>
##  1 01001020100 Alab~ Autau~     1    1912     693      162.            0    79.5
##  2 01001020200 Alab~ Autau~     1    2170     743       51.0           0    41.7
##  3 01001020300 Alab~ Autau~     1    3373    1256      190.            0    50.3
##  4 01001020400 Alab~ Autau~     1    4386    1722      237.            0    24.4
##  5 01001020500 Alab~ Autau~     1   10766    4082      362.            0   120.
##  6 01001020600 Alab~ Autau~     1    3668    1311      264.            0    87.5
##  7 01001020700 Alab~ Autau~     1    2891    1188      109.            0   110.
##  8 01001020801 Alab~ Autau~     0    3081    1074      341.            0   105.
##  9 01001020802 Alab~ Autau~     0   10435    3694     1060.         24.4   365.
## 10 01001020900 Alab~ Autau~     0    5675    2067      630.        242.    391.
## # i 145,385 more rows
## # i 16 more variables: laseniors1share <dbl>, laseniors10share <dbl>,
## #   lasnap1share <dbl>, lasnap10share <dbl>, lasnap10 <dbl>, laseniors20 <dbl>,
## #   laseniorshalf <dbl>, laseniorshalfshare <dbl>, tractseniors <dbl>,
## #   tractsnap <dbl>, year <dbl>, area <chr>, senior_la_n <dbl>,
## #   snap_la_n <dbl>, senior_la_pct <dbl>, snap_la_pct <dbl>
```

```r
dim(combined_data)
```

```
## [1] 145395     25
```

```r
table(combined_data$year)
```

```
##
##  2015  2019
## 72864 72531
```

```r
table(combined_data$area, useNA = "ifany")
```

```
##
##  Rural  Urban
##  35054 110341
```

Explain: After harmonisation, the 2015 and 2019 datasets were combined into a dataset containing 145,395 census-tract observations. Of these, 110,341 were classified as urban and 35,054 as rural, with no missing values in the urban–rural classification.

```r
summary_table <- combined_data %>%
  group_by(year, area) %>%
  summarise(
    n_tracts = n(),

    n_valid_senior =
      sum(!is.na(senior_la_pct)),

    n_valid_snap =
      sum(!is.na(snap_la_pct)),

    mean_senior_pct =
      mean(senior_la_pct, na.rm = TRUE),

    median_senior_pct =
      median(senior_la_pct, na.rm = TRUE),

    mean_snap_pct =
      mean(snap_la_pct, na.rm = TRUE),

    median_snap_pct =
      median(snap_la_pct, na.rm = TRUE),

    .groups = "drop"
  )
#summary_table <- combined_data %>%
#  group_by(year, area) %>%
#  summarise(n_tracts = n(),
#    senior_la_rate =
#      sum(senior_la_n, na.rm = TRUE) /
#      sum(tractseniors, na.rm = TRUE),
#    snap_la_rate =
#      sum(snap_la_n, na.rm = TRUE) /
#      sum(tractsnap, na.rm = TRUE),
#    .groups = "drop"
#  ) %>%

#  mutate(
#    senior_la_pct =
#      senior_la_rate * 100,
#    snap_la_pct =
#      snap_la_rate * 100
# )

summary_table
```

```
## # A tibble: 4 x 9
##    year area  n_tracts n_valid_senior n_valid_snap mean_senior_pct
##   <dbl> <chr>    <int>          <int>        <int>           <dbl>
## 1  2015 Rural    17692          17692        17692            1.68
## 2  2015 Urban    55172          55172        55172            3.37
## 3  2019 Rural    17362           7392         7404            3.96
## 4  2019 Urban    55169          35186        35209            5.32
```

```
## # i 3 more variables: median_senior_pct <dbl>, mean_snap_pct <dbl>,
## #   median_snap_pct <dbl>
```

```
summary_table_report <- summary_table %>%
  mutate(
    across(
      c(
        mean_senior_pct,
        median_senior_pct,
        mean_snap_pct,
        median_snap_pct
      ),
      ~ round(.x, 2)
    )
  )

summary_table_report
```

```
## # A tibble: 4 x 9
##    year area  n_tracts n_valid_senior n_valid_snap mean_senior_pct
##   <dbl> <chr>    <int>          <int>        <int>           <dbl>
## 1  2015 Rural    17692          17692        17692            1.68
## 2  2015 Urban    55172          55172        55172            3.37
## 3  2019 Rural    17362           7392         7404            3.96
## 4  2019 Urban    55169          35186        35209            5.32
## # i 3 more variables: median_senior_pct <dbl>, mean_snap_pct <dbl>,
## #   median_snap_pct <dbl>
```

```
dim(data2015_clean)
```

```
## [1] 72864    25
```

```
names(data2015_clean)
```

```
##  [1] "censustract"        "state"              "county"
##  [4] "urban"              "pop2010"            "ohu2010"
##  [7] "laseniors1"         "laseniors10"        "lasnap1"
## [10] "laseniors1share"    "laseniors10share"   "lasnap1share"
## [13] "lasnap10share"      "lasnap10"           "laseniors20"
## [16] "laseniorshalf"      "laseniorshalfshare" "tractseniors"
## [19] "tractsnap"          "year"               "area"
## [22] "senior_la_n"        "snap_la_n"          "senior_la_pct"
## [25] "snap_la_pct"
```

```
str(data2015_clean)
```

```
## tibble [72,864 x 25] (S3: tbl_df/tbl/data.frame)
##  $ censustract       : chr [1:72864] "01001020100" "01001020200" "01001020300" "01001020400" ...
##  $ state             : chr [1:72864] "Alabama" "Alabama" "Alabama" "Alabama" ...
##  $ county            : chr [1:72864] "Autauga" "Autauga" "Autauga" "Autauga" ...
##  $ urban             : num [1:72864] 1 1 1 1 1 1 1 0 0 0 ...
##  $ pop2010           : num [1:72864] 1912 2170 3373 4386 10766 ...
##  $ ohu2010           : num [1:72864] 693 743 1256 1722 4082 ...
##  $ laseniors1        : num [1:72864] 162 51 190 237 362 ...
##  $ laseniors10       : num [1:72864] 0 0 0 0 0 ...
##  $ lasnap1           : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
##  $ laseniors1share   : num [1:72864] 0.085 0.0235 0.0563 0.0539 0.0336 ...
```

```
##  $ laseniors10share  : num [1:72864] 0 0 0 0 0 ...
##  $ lasnap1share      : num [1:72864] 0.1148 0.0561 0.04 0.0142 0.0294 ...
##  $ lasnap10share     : num [1:72864] 0 0 0 0 0 ...
##  $ lasnap10          : num [1:72864] 0 0 0 0 0 ...
##  $ laseniors20       : num [1:72864] 0 0 0 0 0 0 0 0 0 0 ...
##  $ laseniorshalf     : num [1:72864] 199 139 346 767 840 ...
##  $ laseniorshalfshare: num [1:72864] 0.104 0.0642 0.1026 0.1748 0.078 ...
##  $ tractseniors      : num [1:72864] 221 214 439 904 1126 ...
##  $ tractsnap         : num [1:72864] 112 202 120 82 488 118 218 115 367 394 ...
##  $ year              : num [1:72864] 2015 2015 2015 2015 2015 ...
##  $ area              : chr [1:72864] "Urban" "Urban" "Urban" "Urban" ...
##  $ senior_la_n       : num [1:72864] 162 51 190 237 362 ...
##  $ snap_la_n         : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
##  $ senior_la_pct     : num [1:72864] 8.5 2.35 5.63 5.39 3.36 ...
##  $ snap_la_pct       : num [1:72864] 11.48 5.61 4 1.42 2.94 ...
```

```
summary(data2015_clean)
```

```
##  censustract           state              county
##  Length:72864       Length:72864       Length:72864
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##
##
##
##      urban                  pop2010                 ohu2010
##  Min.   :0.000000000000   Min.   :    0.0000000   Min.   :    0.00000000
##  1st Qu.:1.000000000000   1st Qu.: 2883.7500000   1st Qu.: 1102.00000000
##  Median :1.000000000000   Median : 4002.0000000   Median : 1521.00000000
##  Mean   :0.757191480018   Mean   : 4237.2850516   Mean   : 1601.83756039
##  3rd Qu.:1.000000000000   3rd Qu.: 5323.0000000   3rd Qu.: 2018.00000000
##  Max.   :1.000000000000   Max.   :37452.0000000   Max.   :16043.00000000
##    laseniors1              laseniors10               lasnap1
##  Min.   :    0.000000000   Min.   :   0.0000000000   Min.   :   0.0000000000
##  1st Qu.:    0.000000000   1st Qu.:   0.0000000000   1st Qu.:   0.0000000000
##  Median :  111.152876843   Median :   0.0000000000   Median :  20.1084352526
##  Mean   :  230.558283934   Mean   :  12.0994974777   Mean   :  71.4784359749
##  3rd Qu.:  390.563446517   3rd Qu.:   0.0000000000   3rd Qu.: 103.0891161450
##  Max.   :10225.694634300   Max.   :2530.9999906900   Max.   :1521.2415445500
##  laseniors1share          laseniors10share          lasnap1share
##  Min.   :0.0000000000000   Min.   :0.00000000000000   Min.   :0.0000000000000
##  1st Qu.:0.0000000000000   1st Qu.:0.00000000000000   1st Qu.:0.0000000000000
##  Median :0.0276110682663   Median :0.00000000000000   Median :0.0132343918389
##  Mean   :0.0563605601272   Mean   :0.00413024682724   Mean   :0.0460208855680
##  3rd Qu.:0.0988746660144   3rd Qu.:0.00000000000000   3rd Qu.:0.0673183309857
##  Max.   :1.0000000298000   Max.   :0.57680036250900   Max.   :1.0000000217700
##  lasnap10share              lasnap10
##  Min.   :0.00000000000000   Min.   :    0.00000000
##  1st Qu.:0.00000000000000   1st Qu.:    0.00000000
##  Median :0.00000000000000   Median :    0.00000000
##  Mean   :0.00311638858043   Mean   :    3.71003338017
##  3rd Qu.:0.00000000000000   3rd Qu.:    0.00000000
##  Max.   :0.88888886780400   Max.   :1108.68866587000
##   laseniors20              laseniorshalf            laseniorshalfshare
##  Min.   :   0.00000000000   Min.   :    0.000000000   Min.   :0.0000000000000
```

```
##  1st Qu.:   0.00000000000   1st Qu.:  153.679179371   1st Qu.:0.0424771214540
##  Median :   0.00000000000   Median :  345.999997497   Median :0.0914459654018
##  Mean   :   1.49147507485   Mean   :  395.213484884   Mean   :0.0964164394142
##  3rd Qu.:   0.00000000000   3rd Qu.:  567.160194513   3rd Qu.:0.1374729178860
##  Max.   :2081.12934891000   Max.   :15260.141464200   Max.   :1.0000000298000
##   tractseniors               tractsnap                   year
##  Min.   :     0.000000000   Min.   :   0.000000000   Min.   :2015
##  1st Qu.:   318.000000000   1st Qu.:  69.000000000   1st Qu.:2015
##  Median :   495.000000000   Median : 158.000000000   Median :2015
##  Mean   :   552.645805885   Mean   : 207.022205753   Mean   :2015
##  3rd Qu.:   716.000000000   3rd Qu.: 293.000000000   3rd Qu.:2015
##  Max.   :17271.000000000   Max.   :2152.000000000   Max.   :2015
##      area             senior_la_n                  snap_la_n
##  Length:72864       Min.   :     0.00000000000   Min.   :    0.000000000000
##  Class :character   1st Qu.:     0.00000000000   1st Qu.:    0.000000000000
##  Mode  :character   Median :     5.79194930115   Median :    0.933608777821
##                     Mean   :   123.77540303500   Mean   :   38.252579180900
##                     3rd Qu.:   167.16781171100   3rd Qu.:   37.768104004800
##                     Max.   :10225.69463430000   Max.   :1521.241544550000
##  senior_la_pct              snap_la_pct
##  Min.   :  0.000000000000   Min.   :  0.0000000000000
##  1st Qu.:  0.000000000000   1st Qu.:  0.0000000000000
##  Median :  0.158221080176   Median :  0.0655059568322
##  Mean   :  2.961367271190   Mean   :  2.4835267010000
##  3rd Qu.:  4.125176027870   3rd Qu.:  2.4183350131700
##  Max.   :100.000000000000   Max.   :100.0000021770000
```

Data Cleaning and Missing Data.

```r
dim(data2019_clean)
```

```
## [1] 72531    25
```

```r
names(data2019_clean)
```

```
##  [1] "censustract"        "state"              "county"
##  [4] "urban"              "pop2010"            "ohu2010"
##  [7] "laseniors1"         "laseniors10"        "laseniors1share"
## [10] "laseniors10share"   "lasnap1share"       "lasnap10share"
## [13] "lasnap1"            "lasnap10"           "laseniors20"
## [16] "laseniorshalf"      "laseniorshalfshare" "tractseniors"
## [19] "tractsnap"          "year"               "area"
## [22] "senior_la_n"        "snap_la_n"          "senior_la_pct"
## [25] "snap_la_pct"
```

```r
str(data2019_clean)
```

```
## tibble [72,531 x 25] (S3: tbl_df/tbl/data.frame)
##  $ censustract       : chr [1:72531] "01001020100" "01001020200" "01001020300" "01001020400" ...
##  $ state             : chr [1:72531] "Alabama" "Alabama" "Alabama" "Alabama" ...
##  $ county            : chr [1:72531] "Autauga County" "Autauga County" "Autauga County" "Autauga Cou
##  $ urban             : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
##  $ pop2010           : num [1:72531] 1912 2170 3373 4386 10766 ...
##  $ ohu2010           : num [1:72531] 693 743 1256 1722 4082 ...
##  $ laseniors1        : num [1:72531] 219 127 201 237 362 ...
##  $ laseniors10       : num [1:72531] NA NA NA NA NA ...
##  $ laseniors1share   : num [1:72531] 11.44 5.83 5.96 5.39 3.36 ...
```

```
##  $ laseniors10share  : num [1:72531] NA NA NA NA NA ...
##  $ lasnap1share      : num [1:72531] 13.22 12.95 5.87 1.76 2.04 ...
##  $ lasnap10share     : num [1:72531] NA NA NA NA NA ...
##  $ lasnap1           : num [1:72531] 91.6 96.2 73.7 30.3 83.1 ...
##  $ lasnap10          : num [1:72531] NA NA NA NA NA ...
##  $ laseniors20       : num [1:72531] NA NA NA NA NA NA NA NA NA NA ...
##  $ laseniorshalf     : num [1:72531] 221 214 358 767 840 ...
##  $ laseniorshalfshare: num [1:72531] 11.56 9.86 10.6 17.48 7.8 ...
##  $ tractseniors      : num [1:72531] 221 214 439 904 1126 ...
##  $ tractsnap         : num [1:72531] 102 156 172 98 339 224 390 143 352 340 ...
##  $ year              : num [1:72531] 2019 2019 2019 2019 2019 ...
##  $ area              : chr [1:72531] "Urban" "Urban" "Urban" "Urban" ...
##  $ senior_la_n       : num [1:72531] 219 127 201 237 362 ...
##  $ snap_la_n         : num [1:72531] 91.6 96.2 73.7 30.3 83.1 ...
##  $ senior_la_pct     : num [1:72531] 11.44 5.83 5.96 5.39 3.36 ...
##  $ snap_la_pct       : num [1:72531] 13.22 12.95 5.87 1.76 2.04 ...
```

```
summary(data2019_clean)
```

```
##  censustract           state              county
##  Length:72531       Length:72531       Length:72531
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##      urban                   pop2010                 ohu2010
##  Min.   :0.000000000000   Min.   :    1.00000000   Min.   :    0.00000000
##  1st Qu.:1.000000000000   1st Qu.: 2899.00000000   1st Qu.: 1108.00000000
##  Median :1.000000000000   Median : 4011.00000000   Median : 1525.00000000
##  Mean   :0.760626490742   Mean   : 4256.73902194   Mean   : 1609.19182143
##  3rd Qu.:1.000000000000   3rd Qu.: 5330.50000000   3rd Qu.: 2021.00000000
##  Max.   :1.000000000000   Max.   :37452.00000000   Max.   :16043.00000000
##
##    laseniors1                laseniors10              laseniors1share
##  Min.   :    0.0000000000   Min.   :   0.0000000000   Min.   :  0.00000000000
##  1st Qu.:   73.7881634476   1st Qu.:   7.5835490330   1st Qu.:  1.87904652929
##  Median :  259.2818331850   Median :  51.6980597595   Median :  6.54900459429
##  Mean   :  319.2990643370   Mean   : 111.1584019070   Mean   :  7.82625335695
##  3rd Qu.:  484.0000007010   3rd Qu.: 162.3787189970   3rd Qu.: 12.22684722590
##  Max.   :10348.5798401000   Max.   :2530.9999906900   Max.   :100.00000298000
##  NA's   :19989              NA's   :64765             NA's   :19989
##  laseniors10share         lasnap1share            lasnap10share
##  Min.   : 0.0000000000   Min.   :  0.00000000000   Min.   : 0.0000000000
##  1st Qu.: 0.1969509957   1st Qu.:  0.79378705758   1st Qu.: 0.1106945645
##  Median : 1.5001749784   Median :  3.43137257330   Median : 0.8474116327
##  Mean   : 3.8071163129   Mean   :  5.99596583627   Mean   : 2.6684463901
##  3rd Qu.: 5.3015079945   3rd Qu.:  8.65955330459   3rd Qu.: 3.0099299123
##  Max.   :57.6800362509   Max.   :100.00000199900   Max.   :84.5265517912
##  NA's   :64765           NA's   :19966             NA's   :64666
##     lasnap1                 lasnap10               laseniors20
##  Min.   :   0.0000000000   Min.   :  0.0000000000   Min.   :   0.0000000000
##  1st Qu.:  11.8921063110   1st Qu.:  1.7691137766   1st Qu.:   0.4210301405
##  Median :  52.9257316770   Median : 11.3287673211   Median :  12.4007253413
```

```
##  Mean   :  92.5222140142   Mean   : 31.9297830959   Mean   :  63.2935971552
##  3rd Qu.: 133.5308377880   3rd Qu.: 37.9660582705   3rd Qu.:  72.9691026808
##  Max.   :1582.3107130700   Max.   :995.1762471410   Max.   :2081.1293489100
##  NA's   :19989             NA's   :64765            NA's   :71025
##  laseniorshalf             laseniorshalfshare         tractseniors
##  Min.   :     0.000000000   Min.   :  0.00000000000   Min.   :    0.000000000
##  1st Qu.:   195.009266620   1st Qu.:  5.28589415698   1st Qu.:  320.000000000
##  Median :   372.666346259   Median :  9.76757168611   Median :  497.000000000
##  Mean   :   422.927560834   Mean   : 10.32568201080   Mean   :  555.197112799
##  3rd Qu.:   585.757530171   3rd Qu.: 14.08033594170   3rd Qu.:  718.000000000
##  Max.   :15261.080946900   Max.   :100.00000298000   Max.   :17271.000000000
##  NA's   :4568              NA's   :4568              NA's   :4
##    tractsnap                   year          area
##  Min.   :   0.000000000   Min.   :2019   Length:72531
##  1st Qu.:  67.000000000   1st Qu.:2019   Class :character
##  Median : 152.000000000   Median :2019   Mode  :character
##  Mean   : 201.753181574   Mean   :2019
##  3rd Qu.: 282.000000000   3rd Qu.:2019
##  Max.   :2175.000000000   Max.   :2019
##  NA's   :4
##   senior_la_n                 snap_la_n
##  Min.   :     0.0000000000   Min.   :    0.00000000000
##  1st Qu.:    25.8028090326   1st Qu.:    4.59093786055
##  Median :   125.9808855350   Median :   24.81192350920
##  Mean   :   211.4167894130   Mean   :   61.94091236500
##  3rd Qu.:   308.0129940800   3rd Qu.:   78.62356775790
##  Max.   :10348.5798401000   Max.   :1582.31071307000
##  NA's   :29953              NA's   :29953
##  senior_la_pct             snap_la_pct
##  Min.   :  0.00000000000   Min.   : 0.00000000000
##  1st Qu.:  0.66776057096   1st Qu.: 0.30361097156
##  Median :  3.15722085022   Median : 1.59971655055
##  Mean   :  5.08570513756   Mean   : 4.06176372883
##  3rd Qu.:  7.53046431934   3rd Qu.: 4.86934977853
##  Max.   :100.00000000000   Max.   :85.09091031990
##  NA's   :29953             NA's   :29918
```

```r
colSums(is.na(data2019_clean))
```

```
##        censustract              state             county              urban
##                  0                  0                  0                  0
##            pop2010            ohu2010         laseniors1        laseniors10
##                  0                  0              19989              64765
##    laseniors1share  laseniors10share        lasnap1share       lasnap10share
##              19989              64765              19966              64666
##            lasnap1           lasnap10        laseniors20      laseniorshalf
##              19989              64765              71025               4568
## laseniorshalfshare       tractseniors          tractsnap               year
##               4568                  4                  4                  0
##               area        senior_la_n          snap_la_n     senior_la_pct
##                  0              29953              29953              29953
##        snap_la_pct
##              29918
```

```r
dim(data2015_clean)
```

```
## [1] 72864    25
```

```r
names(data2015_clean)
```

```
##  [1] "censustract"        "state"              "county"
##  [4] "urban"              "pop2010"            "ohu2010"
##  [7] "laseniors1"         "laseniors10"        "lasnap1"
## [10] "laseniors1share"    "laseniors10share"   "lasnap1share"
## [13] "lasnap10share"      "lasnap10"           "laseniors20"
## [16] "laseniorshalf"      "laseniorshalfshare" "tractseniors"
## [19] "tractsnap"          "year"               "area"
## [22] "senior_la_n"        "snap_la_n"          "senior_la_pct"
## [25] "snap_la_pct"
```

```r
str(data2015_clean)
```

```
## tibble [72,864 x 25] (S3: tbl_df/tbl/data.frame)
##  $ censustract       : chr [1:72864] "01001020100" "01001020200" "01001020300" "01001020400" ...
##  $ state             : chr [1:72864] "Alabama" "Alabama" "Alabama" "Alabama" ...
##  $ county            : chr [1:72864] "Autauga" "Autauga" "Autauga" "Autauga" ...
##  $ urban             : num [1:72864] 1 1 1 1 1 1 1 0 0 0 ...
##  $ pop2010           : num [1:72864] 1912 2170 3373 4386 10766 ...
##  $ ohu2010           : num [1:72864] 693 743 1256 1722 4082 ...
##  $ laseniors1        : num [1:72864] 162 51 190 237 362 ...
##  $ laseniors10       : num [1:72864] 0 0 0 0 0 ...
##  $ lasnap1           : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
##  $ laseniors1share   : num [1:72864] 0.085 0.0235 0.0563 0.0539 0.0336 ...
##  $ laseniors10share  : num [1:72864] 0 0 0 0 0 ...
##  $ lasnap1share      : num [1:72864] 0.1148 0.0561 0.04 0.0142 0.0294 ...
##  $ lasnap10share     : num [1:72864] 0 0 0 0 0 ...
##  $ lasnap10          : num [1:72864] 0 0 0 0 0 ...
##  $ laseniors20       : num [1:72864] 0 0 0 0 0 0 0 0 0 0 ...
##  $ laseniorshalf     : num [1:72864] 199 139 346 767 840 ...
##  $ laseniorshalfshare: num [1:72864] 0.104 0.0642 0.1026 0.1748 0.078 ...
##  $ tractseniors      : num [1:72864] 221 214 439 904 1126 ...
##  $ tractsnap         : num [1:72864] 112 202 120 82 488 118 218 115 367 394 ...
##  $ year              : num [1:72864] 2015 2015 2015 2015 2015 ...
##  $ area              : chr [1:72864] "Urban" "Urban" "Urban" "Urban" ...
##  $ senior_la_n       : num [1:72864] 162 51 190 237 362 ...
##  $ snap_la_n         : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
##  $ senior_la_pct     : num [1:72864] 8.5 2.35 5.63 5.39 3.36 ...
##  $ snap_la_pct       : num [1:72864] 11.48 5.61 4 1.42 2.94 ...
```

```r
summary(data2015_clean)
```

```
##  censustract           state              county
##  Length:72864       Length:72864       Length:72864
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##
##
##
##      urban                    pop2010                 ohu2010
##  Min.   :0.000000000000   Min.   :     0.0000000   Min.   :     0.00000000
```

```
##  1st Qu.:1.000000000000   1st Qu.: 2883.7500000   1st Qu.: 1102.00000000
##  Median :1.000000000000   Median : 4002.0000000   Median : 1521.00000000
##  Mean   :0.757191480018   Mean   : 4237.2850516   Mean   : 1601.83756039
##  3rd Qu.:1.000000000000   3rd Qu.: 5323.0000000   3rd Qu.: 2018.00000000
##  Max.   :1.000000000000   Max.   :37452.0000000   Max.   :16043.00000000
##    laseniors1              laseniors10               lasnap1
##  Min.   :     0.000000000   Min.   :   0.0000000000   Min.   :   0.0000000000
##  1st Qu.:     0.000000000   1st Qu.:   0.0000000000   1st Qu.:   0.0000000000
##  Median :   111.152876843   Median :   0.0000000000   Median :  20.1084352526
##  Mean   :   230.558283934   Mean   :  12.0994974777   Mean   :  71.4784359749
##  3rd Qu.:   390.563446517   3rd Qu.:   0.0000000000   3rd Qu.: 103.0891161450
##  Max.   :10225.694634300   Max.   :2530.9999906900   Max.   :1521.2415445500
##  laseniors1share          laseniors10share          lasnap1share
##  Min.   :0.0000000000000   Min.   :0.00000000000000   Min.   :0.0000000000000
##  1st Qu.:0.0000000000000   1st Qu.:0.00000000000000   1st Qu.:0.0000000000000
##  Median :0.0276110682663   Median :0.00000000000000   Median :0.0132343918389
##  Mean   :0.0563605601272   Mean   :0.00413024682724   Mean   :0.0460208855680
##  3rd Qu.:0.0988746660144   3rd Qu.:0.00000000000000   3rd Qu.:0.0673183309857
##  Max.   :1.0000000298000   Max.   :0.57680036250900   Max.   :1.0000000217700
##  lasnap10share              lasnap10
##  Min.   :0.00000000000000   Min.   :   0.00000000000
##  1st Qu.:0.00000000000000   1st Qu.:   0.00000000000
##  Median :0.00000000000000   Median :   0.00000000000
##  Mean   :0.00311638858043   Mean   :   3.71003338017
##  3rd Qu.:0.00000000000000   3rd Qu.:   0.00000000000
##  Max.   :0.88888886780400   Max.   :1108.68866587000
##   laseniors20              laseniorshalf             laseniorshalfshare
##  Min.   :   0.00000000000   Min.   :     0.000000000   Min.   :0.0000000000000
##  1st Qu.:   0.00000000000   1st Qu.:   153.679179371   1st Qu.:0.0424771214540
##  Median :   0.00000000000   Median :   345.999997497   Median :0.0914459654018
##  Mean   :   1.49147507485   Mean   :   395.213484884   Mean   :0.0964164394142
##  3rd Qu.:   0.00000000000   3rd Qu.:   567.160194513   3rd Qu.:0.1374729178860
##  Max.   :2081.12934891000   Max.   :15260.141464200   Max.   :1.0000000298000
##   tractseniors              tractsnap                  year
##  Min.   :     0.000000000   Min.   :   0.000000000   Min.   :2015
##  1st Qu.:   318.000000000   1st Qu.:  69.000000000   1st Qu.:2015
##  Median :   495.000000000   Median : 158.000000000   Median :2015
##  Mean   :   552.645805885   Mean   : 207.022205753   Mean   :2015
##  3rd Qu.:   716.000000000   3rd Qu.: 293.000000000   3rd Qu.:2015
##  Max.   :17271.000000000   Max.   :2152.000000000   Max.   :2015
##      area             senior_la_n                 snap_la_n
##  Length:72864       Min.   :    0.00000000000   Min.   :   0.000000000000
##  Class :character   1st Qu.:    0.00000000000   1st Qu.:   0.000000000000
##  Mode  :character   Median :    5.79194930115   Median :   0.933608777821
##                     Mean   :  123.77540303500   Mean   :  38.252579180900
##                     3rd Qu.:  167.16781171100   3rd Qu.:  37.768104004800
##                     Max.   :10225.69463430000   Max.   :1521.241544550000
##  senior_la_pct              snap_la_pct
##  Min.   :  0.000000000000   Min.   :  0.0000000000000
##  1st Qu.:  0.000000000000   1st Qu.:  0.0000000000000
##  Median :  0.158221080176   Median :  0.0655059568322
##  Mean   :  2.961367271190   Mean   :  2.4835267010000
##  3rd Qu.:  4.125176027870   3rd Qu.:  2.4183350131700
##  Max.   :100.000000000000   Max.   :100.0000021770000
```

```r
colSums(is.na(data2015_clean))
```

```
##        censustract               state              county               urban
##                  0                   0                   0                   0
##            pop2010             ohu2010          laseniors1         laseniors10
##                  0                   0                   0                   0
##            lasnap1    laseniors1share    laseniors10share        lasnap1share
##                  0                   0                   0                   0
##       lasnap10share            lasnap10         laseniors20       laseniorshalf
##                  0                   0                   0                   0
## laseniorshalfshare        tractseniors           tractsnap                year
##                  0                   0                   0                   0
##               area         senior_la_n           snap_la_n      senior_la_pct
##                  0                   0                   0                   0
##       snap_la_pct
##                  0
```

```r
#summary(data2015_clean$snap_la_rate)
#summary(data2019_clean$snap_la_rate)
```

```r
#summary(data2015_clean$senior_la_rate)
#summary(data2019_clean$senior_la_rate)
```

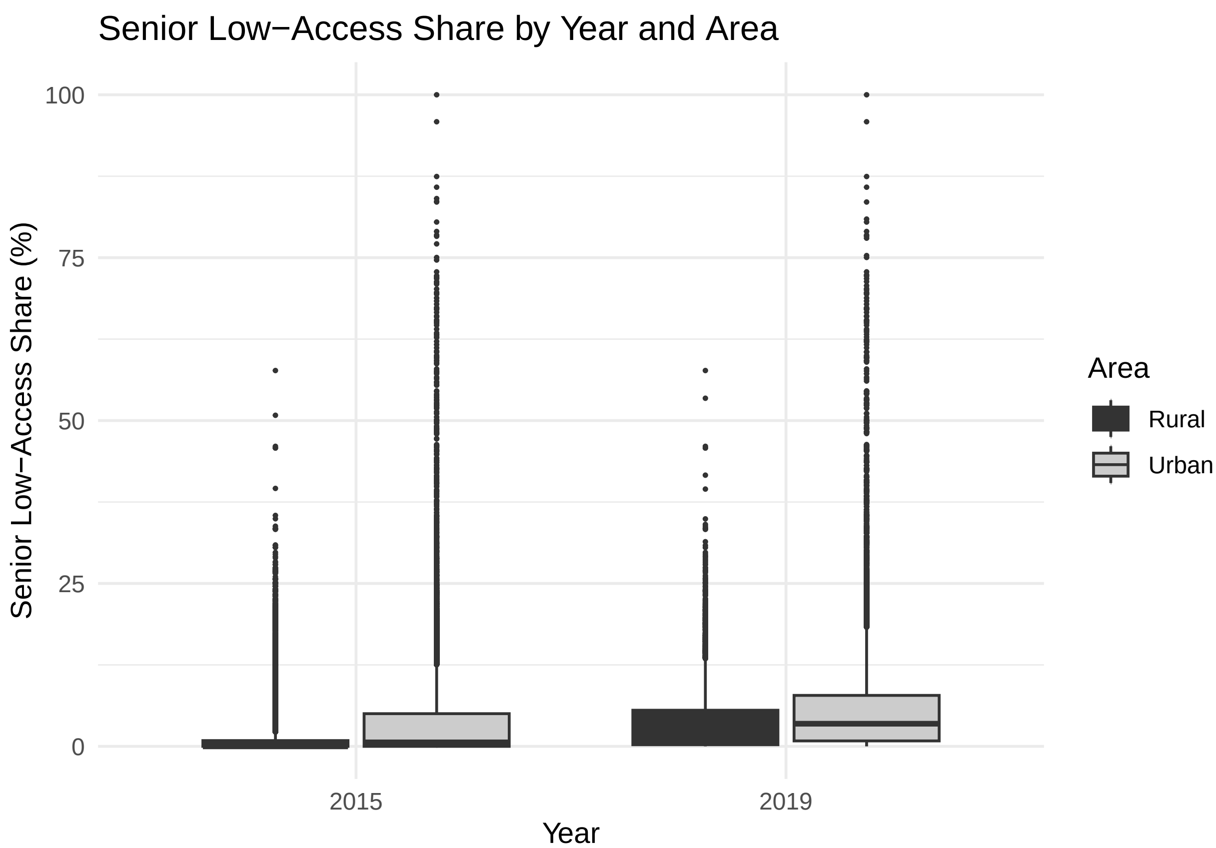
```r
#data2015_clean %>%
#  summarise(
#    n_over_1 = sum(snap_la_rate > 1, na.rm = TRUE),
#    max_rate = max(snap_la_rate, na.rm = TRUE)
#  )

#data2019_clean %>%
#  summarise(
#    n_over_1 = sum(snap_la_rate > 1, na.rm = TRUE),
#    max_rate = max(snap_la_rate, na.rm = TRUE)
#  )
```

```r
#data2015_clean %>%
#  filter(snap_la_rate > 1) %>%
#  select(
#    censustract,
#    urban,
#    tractsnap,
#    lasnap1,
#    lasnap10,
#    snap_la_n,
#    snap_la_rate
#  ) %>%
#  head(20)

#data2019_clean %>%
#  filter(snap_la_rate > 1) %>%
#  select(
#    censustract,
#    urban,
#    tractsnap,
#    lasnap1,
```

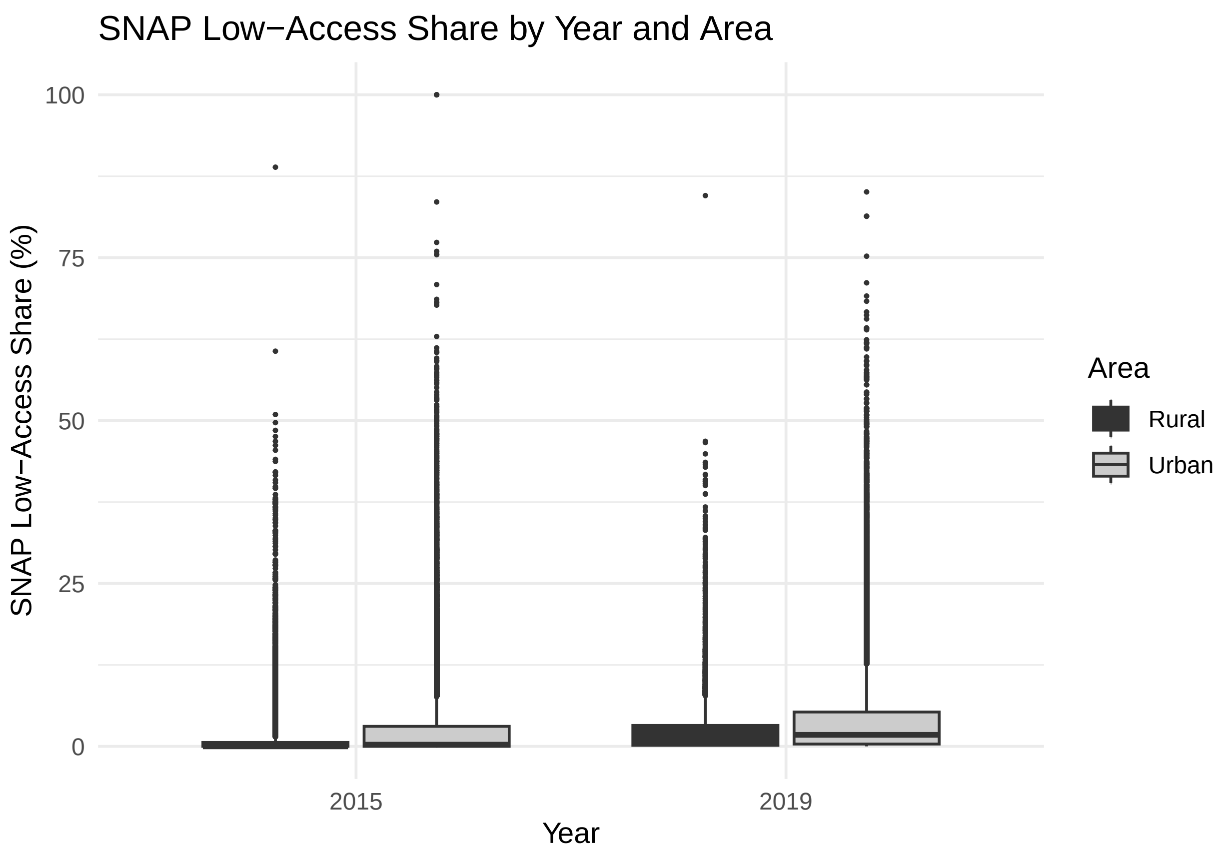
```r
#    lasnap10,
#    snap_la_n,
#    snap_la_rate
#  ) %>%
#  head(20)
```

```r
#data2015_clean %>%
#  summarise(
#    n_numerator_gt_denominator =
#      sum(snap_la_n > tractsnap, na.rm = TRUE)
#  )

#data2019_clean %>%
#  summarise(
#    n_numerator_gt_denominator =
#      sum(snap_la_n > tractsnap, na.rm = TRUE)
#  )
```

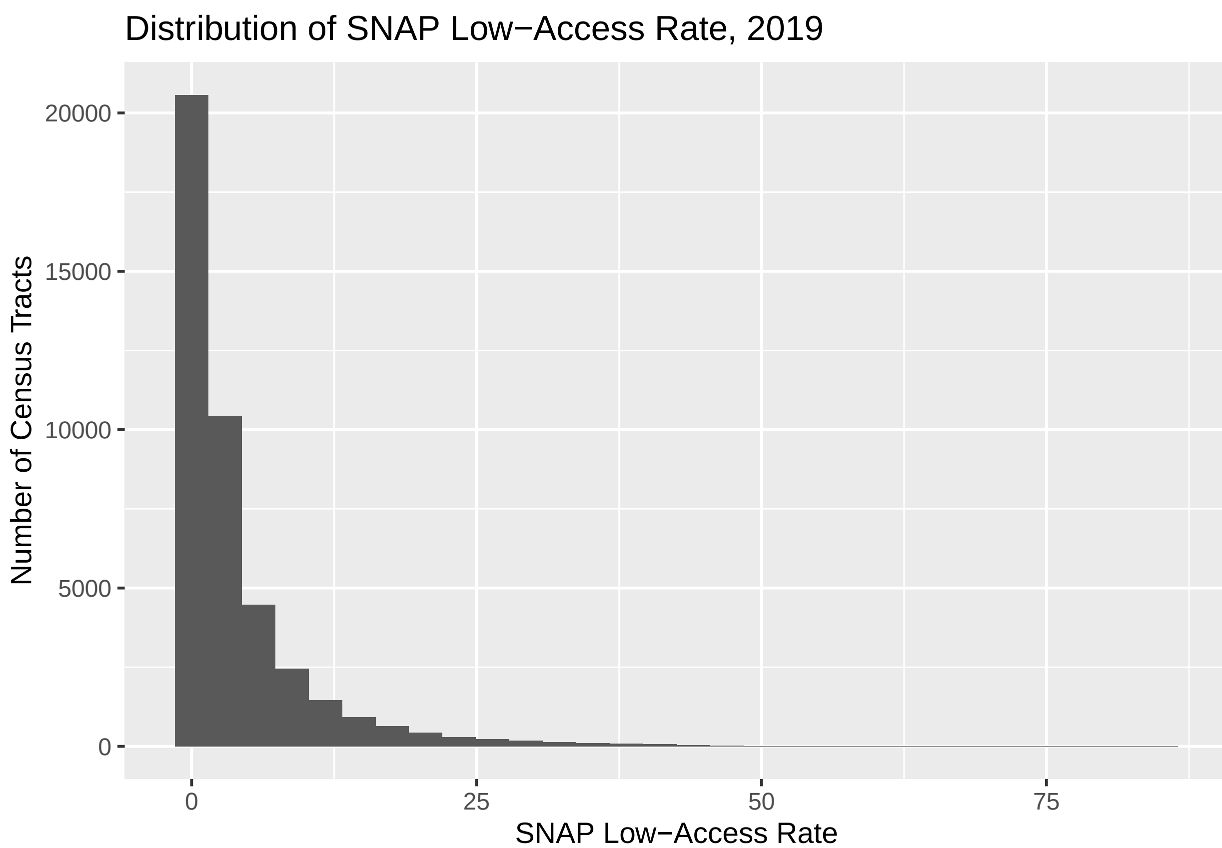
```r
# Senior Boxplot
ggplot(
  combined_data,
  aes(
    x = factor(year),
    y = senior_la_pct,
    fill = area
  )
) +
  geom_boxplot(
    outlier.size = 0.3,
    na.rm = TRUE
  ) +
  scale_fill_grey() +
  labs(
    title = "Senior Low-Access Share by Year and Area",
    x = "Year",
    y = "Senior Low-Access Share (%)",
    fill = "Area"
  ) +
  theme_minimal()
```



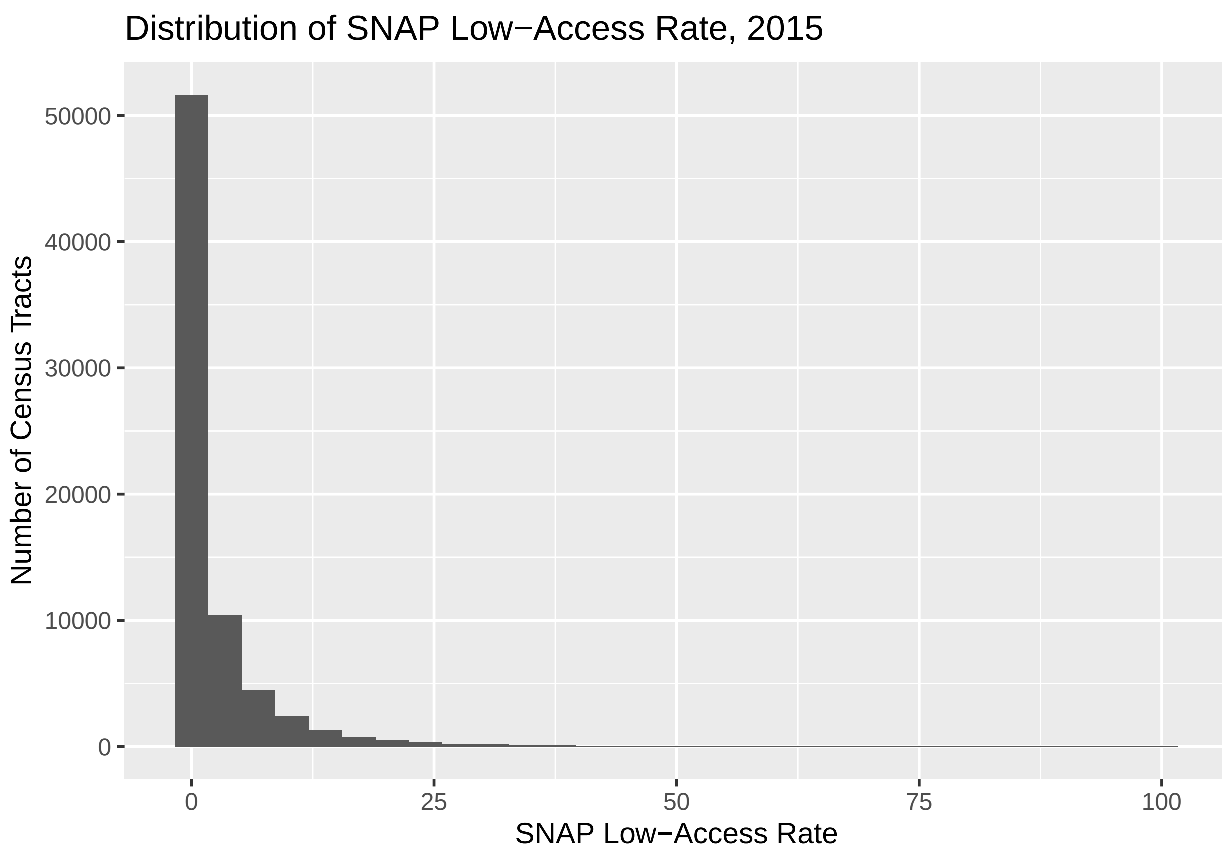

```r
# SNAP boxplot
ggplot(
  combined_data,
  aes(
    x = factor(year),
    y = snap_la_pct,
    fill = area
  )
) +
  geom_boxplot(
    outlier.size = 0.3,
    na.rm = TRUE
  ) +
  scale_fill_grey() +
  labs(
    title = "SNAP Low-Access Share by Year and Area",
    x = "Year",
    y = "SNAP Low-Access Share (%)",
    fill = "Area"
  ) +
  theme_minimal()
```



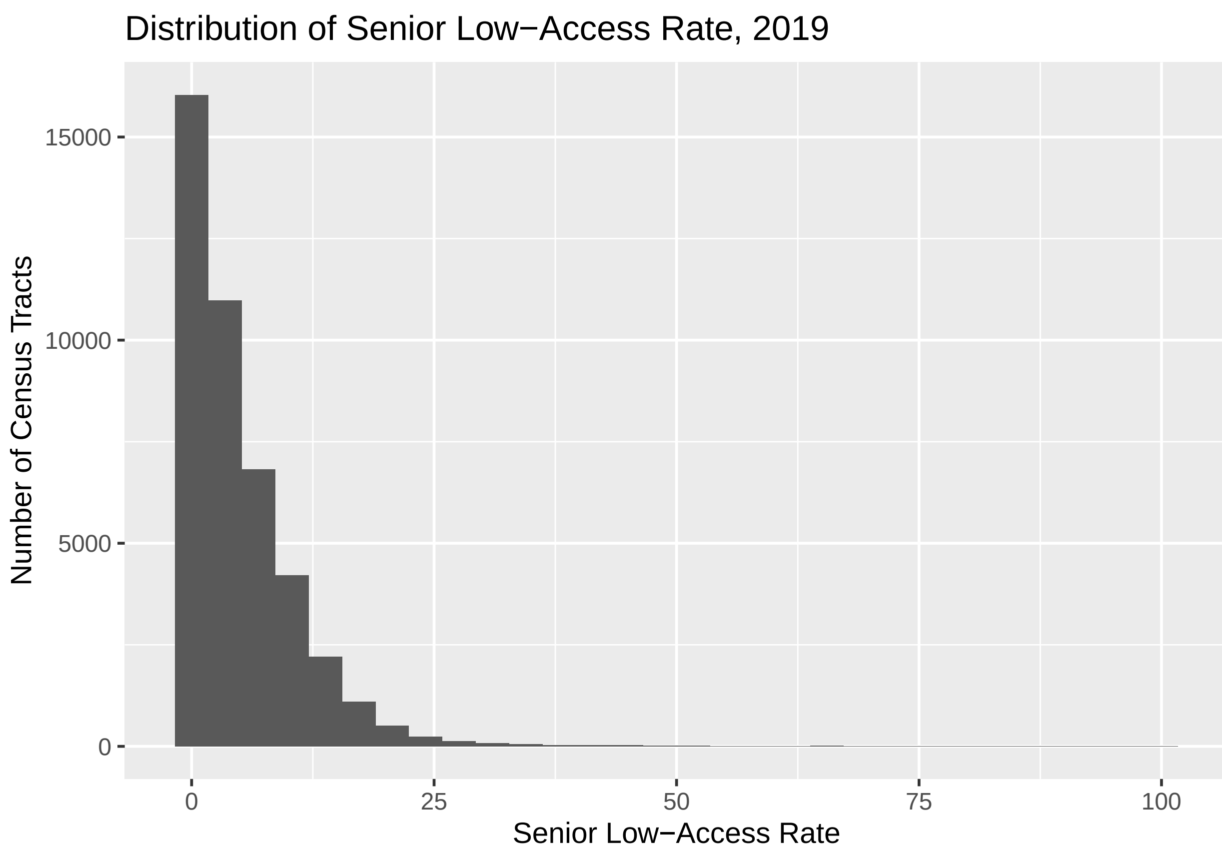

```r
library(ggplot2)
ggplot(data2019_clean, aes(x = snap_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Distribution of SNAP Low-Access Rate, 2019",
    x = "SNAP Low-Access Rate",
    y = "Number of Census Tracts"
  )
```



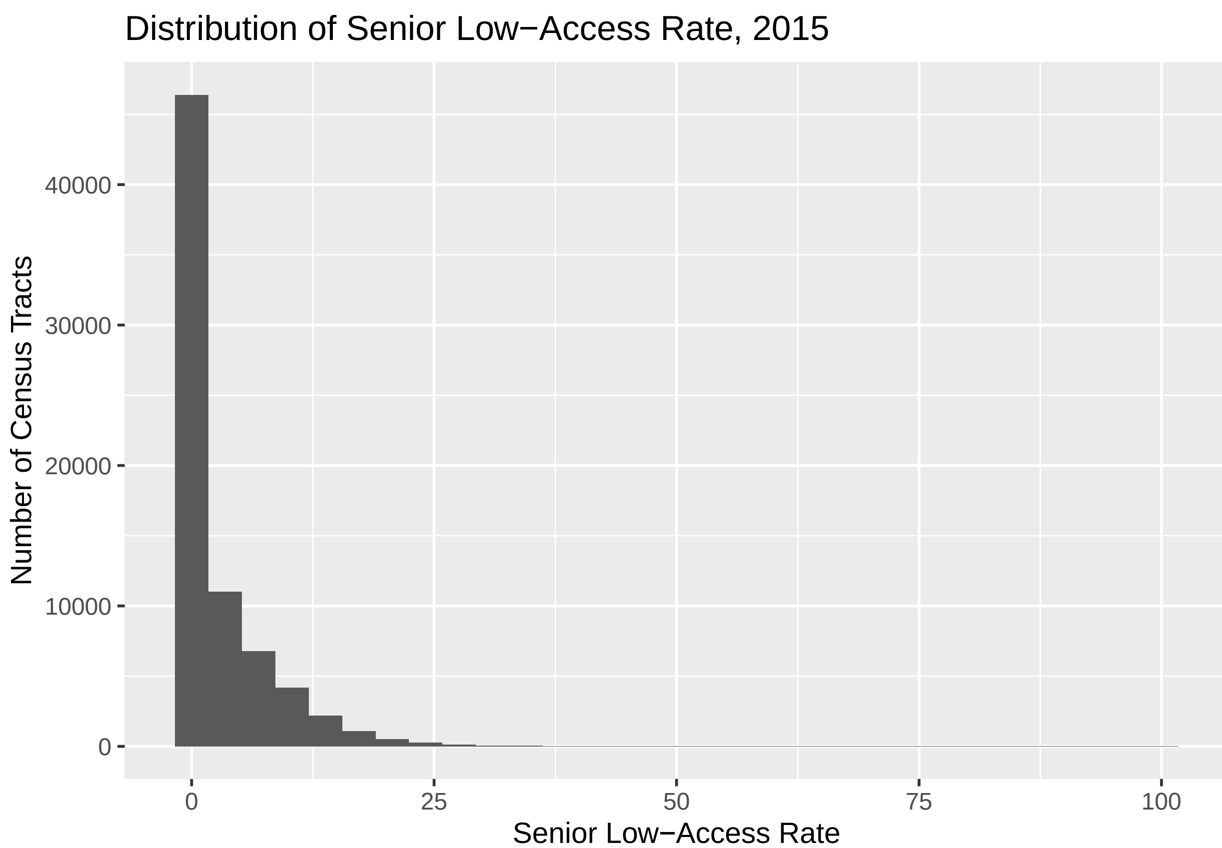

```r
library(ggplot2)
ggplot(data2015_clean, aes(x = snap_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Distribution of SNAP Low-Access Rate, 2015",
    x = "SNAP Low-Access Rate",
    y = "Number of Census Tracts"
  )
```



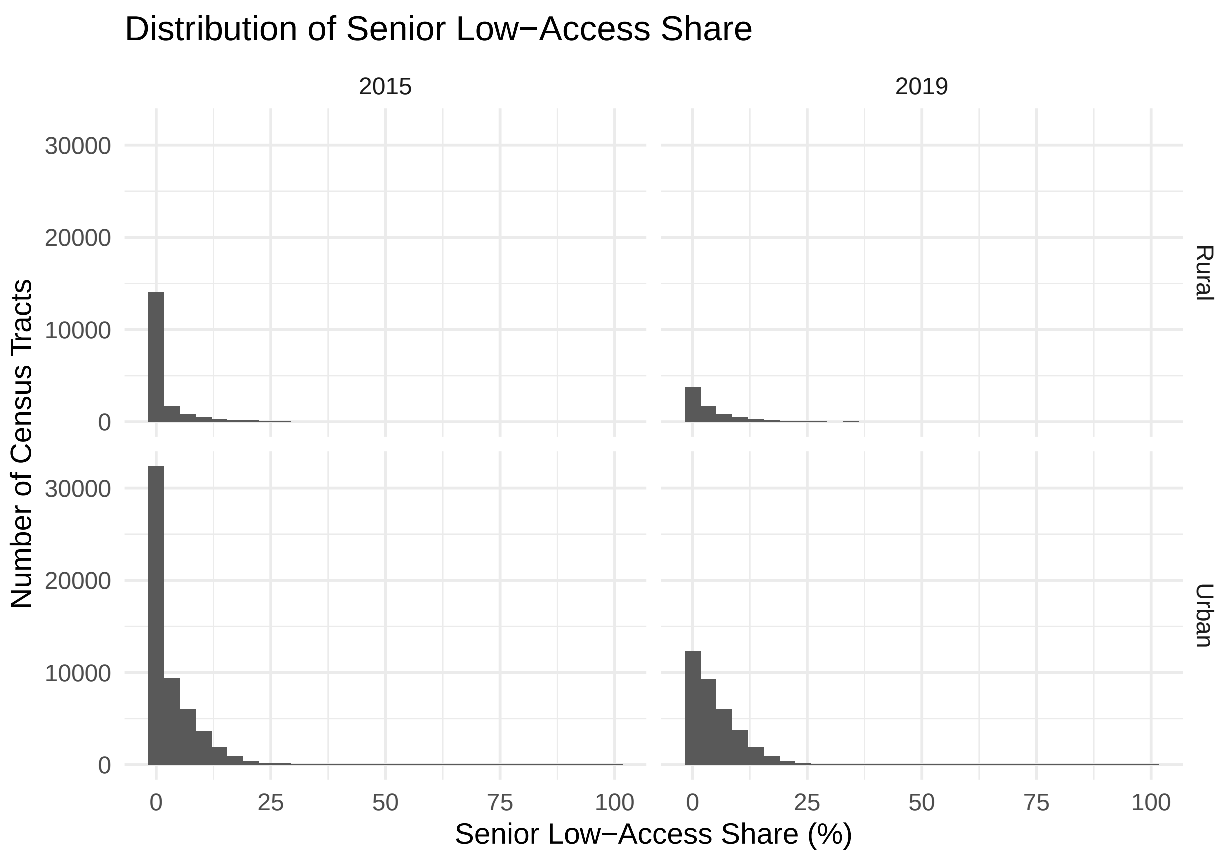

```
library(ggplot2)
ggplot(data2019_clean, aes(x = senior_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Distribution of Senior Low-Access Rate, 2019",
    x = "Senior Low-Access Rate",
    y = "Number of Census Tracts"
  )
```



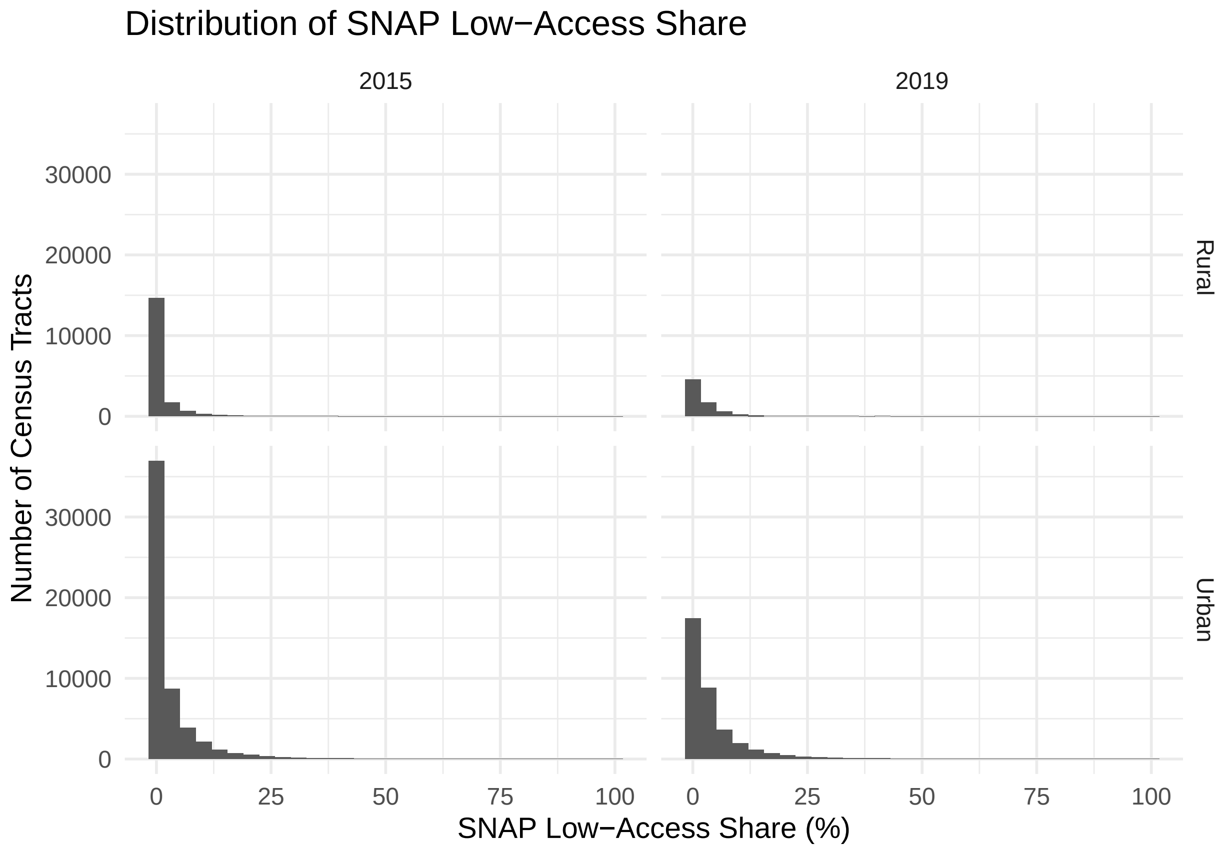

```r
ggplot(data2015_clean, aes(x = senior_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Distribution of Senior Low-Access Rate, 2015",
    x = "Senior Low-Access Rate",
    y = "Number of Census Tracts"
  )
```




```r
ggplot(
  combined_data,
  aes(x = senior_la_pct)
) +
  geom_histogram(
    bins = 30,
    na.rm = TRUE
  ) +
  facet_grid(area ~ year) +
  labs(
    title = "Distribution of Senior Low-Access Share",
    x = "Senior Low-Access Share (%)",
    y = "Number of Census Tracts"
  ) +
  theme_minimal()
```




```r
ggplot(
  combined_data,
  aes(x = snap_la_pct)
) +
  geom_histogram(
    bins = 30,
    na.rm = TRUE
  ) +
  facet_grid(area ~ year) +
  labs(
    title = "Distribution of SNAP Low-Access Share",
    x = "SNAP Low-Access Share (%)",
    y = "Number of Census Tracts"
  ) +
  theme_minimal()
```


Distribution of SNAP Low−Access Share
2015
2019
Rural
Urban
Number of Census Tracts
0
10000
20000
30000
0
25
50
75
100
SNAP Low−Access Share (%)